Multi-Nodal Optimisation of Energy Storage Requirements for 100% Renewable Energy Grid: A South Australian Case Study

Harrison Payne[a], Patrick Moriarty[a], Ali Pourmousavi Kani[a,*], Wen Soong[a], Kevin Peter[b]

[a]*School of Electrical and Electronic Engineering, University of Adelaide North Terrace, Adelaide, SA 5005, Australia*
[b]*Network Planning, ElectraNet East Terrace, Adelaide, SA 5000, Australia*

Abstract

The transition to 100% renewable energy systems presents significant challenges in maintaining grid reliability, particularly in systems with large amounts of variable renewable energy sources. This study develops a comprehensive multi-nodal optimisation framework to assess the long term storage requirements of 100% renewable energy grids under conservative operating assumptions. Using South Australia as a gigawatt-scale case study, the developed mixed-integer linear program is used to assess the storage requirement across seven regions incorporating transmission constraints, DC power flow with linearised losses, multiple storage technologies with temporal classifications, and correlated renewable generation profiles. Under conservative operating conditions, the results indicate a maximum storage requirement of 7.28 GW and 644 GWh hours across the investigation period (2045-2051). Winter is revealed as the most critical season for storage sizing, with widespread renewable energy droughts occurring across geographically correlated regions and therefore limiting the effectiveness of interstate support. Sensitivity analyses demonstrate that modelling more realistic operating scenarios, such as including modest amounts of dispatchable generation (1%-3% of total annual demand), can reduce storage requirements by up to 78%. In contrast, the importance of considering multi-year climate data is demonstrated with storage sizing varying by more than 500 GWh between weather years. Across all analyses, the median requirement of South Australia is 4.27 GW and 162 GWh. The modelling framework and subsequent findings provide valuable insights for policymakers and grid planners working towards the goal of a 100% renewable energy grid.

Keywords: Energy storage, Renewable energy, Grid optimisation, Mixed-integer linear programming, Power system planning, South Australia

*Corresponding author
Email addresses: harrison.payne@adelaide.edu.au (Harrison Payne), patrick.moriarty@adelaide.edu.au (Patrick Moriarty), ali.pourmousavi@adelaide.edu.au (Ali Pourmousavi Kani), wen.soong@adelaide.edu.au (Wen Soong), peter.kevin@electranet.com.au (Kevin Peter)

Nomenclature

Sets and Indices

- $\mathcal{R}$ Set of local electricity grid nodes
- $\mathcal{S}$ Set of neighbouring electricity grid nodes
- $\mathcal{N}$ Set of all nodes ($\mathcal{R} \cup \mathcal{S}$)
- $\mathcal{B}$ Set of storage types (lithium-ion, PHES, etc.)
- $\mathcal{H}$ Set of all temporal storage classifications (short-, medium-, long-term)
- $\mathcal{G}$ Set of VRE generation technologies
- $\mathcal{T}$ Set of time periods
- $\mathcal{L}$ Set of distinct transmission lines ij, $i < j$
- r Local node index
- s Neighbouring node index
- n All nodes index
- b Storage type index
- h Storage temporal classification index
- g Generation technology index
- t Time period index
- ij Transmission line connecting nodes i and j (if one exists)

Decision Variables

- $E_{r,b,h,t}$ Energy stored at local node r, storage type b, temporal classification h, at time t (pu)
- $E^{\max}_{r,b,h}$ Maximum energy capacity for local node r, storage type b, temporal classification h (pu)
- $P^{\max}_{r,b,h}$ Maximum power capacity for local node r, storage type b, temporal classification h (pu)
- $P_{\text{chrg},r,b,h,t}$ Charging power for local node r, storage type b, temporal classification h, at time t (pu)
- $P_{\text{disc},r,b,h,t}$ Discharging power for local node r, storage type b, temporal classification h, at time t (pu)
- $P_{I,n,t}$ Net flow into node n at time t (pu)
- $\theta_{n,t}$ Voltage angle at node n at time t (radians)
- $P_{pos,ij,t}$ Positive composition of transmission line ij power flow, at time t (pu)
- $P_{neg,ij,t}$ Negative composition of transmission line ij power flow, at time t (pu)
- $P_{\text{curt},n,t}$ Curtailed generation at node n at time t (pu)
- $P_{\text{DSP},r,t}$ Demand-side participation at local node r at time t (pu)
- $\text{USE}_{r,t}$ Unserved energy at local node r at time t (pu)
- $\alpha_{r,b,h,t}$ Binary variable to ensure charging and discharging does not occur simultaneously (0,1)

Parameters

S_{base}	Three phase base power (GVA)
$G_{r,t}$	Forecast renewable generation at local node r at time t (pu)
$D_{r,t}$	Forecast electricity demand at local node r at time t (pu)
$C_{E,b,h}$	Energy capacity cost for storage type b of temporal classification h (pu adjusted)
$C_{P,b,h}$	Power capacity cost for storage type b of temporal classification h (pu adjusted)
$C_{O,b,h}$	Present value of operating expenditure over the analysis period for storage type b temporal classification h (pu adjusted)
$C_{A,O,b}$	Annual operating expenditure for storage type b of temporal classification h (pu adjusted)
$PVAF$	Present value annuity factor
dsc	Capital discount rate
N	Horizon which investments are analysed
$G_{r,g,t}$	Utility scale generation at local node r, of generation technology g, at time t (pu)
$\eta_{c,b,h}$	Charging efficiency for storage type b of temporal classification h (%)
$\eta_{d,b,h}$	Discharging efficiency for storage type b of temporal classification h (%)
$P_{ij,t}^{\max}$	Thermal limit of transmission line ij and time t (pu)
Δt	Time interval duration (hours)
X_{ij}	Reactance of transmission line ij (pu)
B_{ij}	$1/X_{ij}$ for transmission line line ij (pu)
θ_{ref}	Reference node voltage angle $\theta_{ref} = 0$ (radians)
β	Transmission line linear loss factor (%)
$P_{DSP}^{\max}$	Demand-side participation power limit (pu)
$E_{DSP}^{\max}$	Demand-side participation energy limit (pu)
hrs_{DSP}	Demand-side participation equivalent number of daily hours of operation (hours)
$\mathrm{USE}^{\max}$	Unserved energy power limit (pu)
$\mathrm{USE}_{\%}$	Annual unserved energy as percentage of total forecast demand (%)
$GS_{s,t}$	Forecast surplus generation in neighbouring node s at time t (pu)
M	Big-M parameter for binary constraints
N_{cycles}	Number of equivalent cycles a battery storage is constrained by
$E_{r,b,h}^{lim}$	Build energy capacity limit at node r, storage type b, of temporal classification h.
$P_{r,b,h}^{lim}$	Build power capacity limit at node r, storage type b, of temporal classification h.

Acronyms

AEMO	Australian Energy Market Operator
CDP	Candidate Development Path
CY	Calendar Year (1st January to 31st December)
DSP	Demand-Side Participation
FY	Financial Year (1st July to 30th June)
GW	Gigawatt
GWh	Gigawatt hour
ISP	Integrated System Plan
LIB	Lithium-Ion Battery
MW	Megawatt
MWh	Megawatt hour
NEM	National Electricity Market
ODP	Optimal Development Path
OPF	Optimal Power Flow
PHES	Pumped Hydro Energy Storage
PtH_2tP	Power-To-Hydrogen-To-Power
REZ	Renewable Energy Zone
SA	South Australia
USE	Unserved Energy
VRE	Variable Renewable Energy

1. Introduction

Electricity grids around the world are undergoing an unprecedented transformation as they transition from fossil fuel dependency towards renewable energy sources. Countries such as Norway, Brazil, and Iceland exemplify this transition and have achieved or are approaching an electricity grid supplied almost entirely by renewable energy [1]. However, these countries have benefited from an abundance of dispatchable hydropower resources. In contrast, other regions including Denmark, Lithuania, and parts of Australia such as South Australia (SA) are more reliant on variable renewable energy (VRE) sources to drive their energy transformation [2].

The transition to net 100% renewable energy, the scenario in which the annual electricity produced from renewable sources in a grid equals or exceeds the total electricity consumption of the grid, and beyond to real 100% renewable energy (RE100), the scenario in which the consumption of electricity is met exclusively by renewable sources at all times, presents complex challenges in maintaining grid reliability and security. Increasing the proportion of VRE generation fundamentally alters the operational characteristics of the electricity system, as the system operators have substantially less control over the balance of supply and demand compared to traditional fossil fuel generators or dispatchable sources of renewable generation [3]. Weather variability and uncertainty, which directly determine wind and solar production [3], create periods of significant renewable energy shortfalls. These shortfalls must be compensated for with adequate energy storage capacity in the case of RE100. This compensation is necessary to maintain reliability, ensuring that only a small percentage of customer demand goes unserved in any given year according to reliability standards.

Critical weather events, including wind droughts (prolonged periods of low wind speeds [4]), extended cloud coverage, and Dunkelflaute conditions (minimal sunshine and wind for extended periods [5]), pose particular risks to electricity networks with a high penetration of renewable energy. These phenomena, combined with multi-year climate patterns such as the El Niño-Southern Oscillation, can significantly impact renewable generation and consequently alter energy storage requirements [6]. The challenge is further compounded by the unique geographical and topographic constraints of certain regions, which can limit the feasibility of certain storage technologies, particularly large-scale pumped hydro energy storage (PHES). Small-scale PHES also face challenges as they cannot leverage economies of scale [7]. In contrast, regions with these limitations are more attractive for alternative storage solutions such as battery storage and emerging technologies such as power-to-hydrogen-to-power (PtH_2tP) systems.

This study aims to address the fundamental question: *"How much energy storage is required for a 100% renewable energy grid while satisfying grid reliability requirements?"* Answering this question requires a sophisticated modelling framework that captures locational context, weather patterns, and technological constraints while remaining flexible enough to be applied to different electricity grids. Crucially, such modelling must ensure that electricity grids continue to meet reliability standards and remain affordable to consumers.

1.1. Literature Review

As renewable energy uptake has increased, so has the literature on the storage requirements of a RE100 grid. Although the objectives and scope of each study differ, three main themes emerge. The first theme is the importance of model flexibility and the ability to capture the spatial dynamics of the region being examined, as the availability of natural resources and the existing network infrastructure can significantly influence the required storage. Furthermore, ensuring that the influence of weather events and multi-year climate patterns were incorporated was another concern, as exploring a single year's worth of data cannot effectively capture the risk of renewable energy droughts. Lastly, the energy storage implementations used and how the storage implementations were temporally categorised played a key role in the estimation of the storage requirements. The subsequent sections explore each of these themes in detail.

1.1.1. Importance of Framework Flexibility

The energy storage requirements for RE100 grids are highly dependent on regional characteristics, which makes location-specific analysis essential. Existing studies have focused predominantly on broader geographical contexts. For example, The Royal Society [8] provides an assessment of large-scale storage technologies and their role in the decarbonisation of Great Britain's energy sector. Furthermore, Ruhnau and Qvist [9] determine the storage requirements for Germany under renewable energy droughts over multiple reference years in the RE100 scenario. In contrast, Jorgenson et al. [10] examine how the widespread adoption of storage in the United States will affect the operational characteristics of power systems. At a regional scale, Bussar et al. [11] analyse the storage demand in the European electricity grid transitioning to 100% renewable energy. Both Schönfisch et al. [12] and the International Energy Agency [13] examine the global scenario with Schönfisch et al. modelling future storage demand from the present to the year 2050 and the International Energy Agency examining the the future needs of energy systems under different policy scenarios. Although these studies provide foundational knowledge, their direct transferability to other electricity systems is often limited. This is primarily due to fundamental assumptions about the availability of renewable resources, the feasibility of storage technologies, and weather patterns, which may not be applicable to different power systems.

Another consideration is the level of abstraction used in the modelling. For example, academic studies focused on Australia have primarily examined scenarios throughout the National Electricity Market (NEM) rather than SA-specific requirements. Blakers et al. [14] estimated 17 gigawatts (GW) and 450 gigawatt hours (GWh) of storage required for the entire NEM using PHES as the primary storage technology, while Shaikh et al. [15] determined 15 GW/227 GWh through copper plate analysis[1]. Multi-nodal NEM analysis by Shaikh et al. [16] suggested that SA would require 1.15 GW/33.6 GWh as part of the interconnected system, while Lu et al. [17] estimated significantly higher SA requirements of 4 GW/121 GWh, although focusing only on PHES technology.

However, such abstractions tend to underestimate the storage requirement for individual regions for two reasons. First, large scale NEM models simplify or omit many operational constraints including granular intra-regional transmission line limits, and spatial load and generation diversity in order to reduce the computational complexity of modelling hundreds to thousands of nodes. These simplifications assume a high

[1] The idealised version of an electricity grid, whereby the connection of all generators and loads is facilitated by zero impedance conductors to a single node.

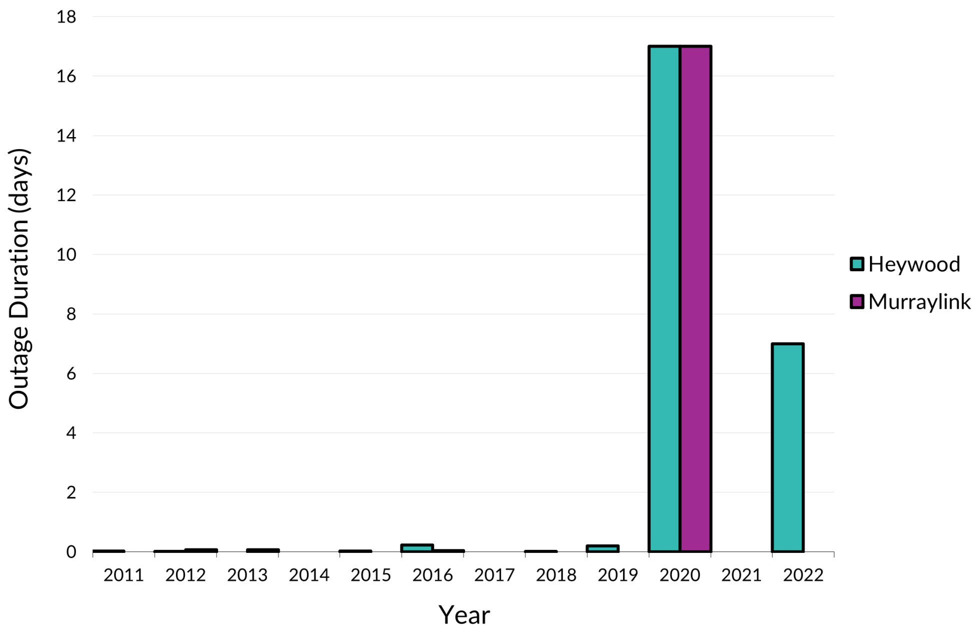

Figure 1: Annual interconnector outage duration in South Australia.

degree of spatial and temporal diversity across the power system. Second, the most challenging operational conditions do not typically occur when a region is connected to another, but rather during periods of islanded or partially islanded operation. In these conditions, inter-regional support is unavailable or limited, meaning that local storage must independently compensate for VRE deficits. Figure 1 illustrates the number of hours per year that SA has been either islanded or partially islanded from the rest of the NEM. Detailed accounts of the events are provided in Appendix A. Furthermore, the aforementioned common incompatibilities are summarised in Table 1 together with a discussion of why their transferability to other power systems is often limited.

Modelling limitations also present themselves on the industry side. The Australian Electricity Market Operator (AEMO) (the overarching planning body in Australia), SA Power Networks (the distribution network service provider) and ElectraNet (the transmission network service provider) all publish industry planning reports, [7], [20], and [21], respectively. However, these generally do not consider RE100 and their modelling inputs and assumptions parameters are generally not disclosed. AEMO releases the most comprehensive data of the three, but there are still limitations.

1.1.2. Weather Events and Climate Pattern Influences

The dependence of renewable generation on weather conditions necessitates a comprehensive understanding of the meteorological impacts on storage requirements. Short-term weather variations significantly affect renewable output, with cloud cover reducing solar irradiance by more than 60% in seconds [22], and low pressure systems creating periods of minimal wind generation [23]. Extreme weather events, including bushfires and dust storms, can reduce solar generation by 15% to 45% during their occurrence [24], demonstrating the critical importance of accounting for such events in storage planning.

Climate modes including El Niño-Southern Oscillation and La Niña significantly influence renewable generation patterns, with studies showing up to 10% variation in winter solar generation [25] and substantial impacts on wind generation patterns [26]. These inter-annual climate variations highlight the need for energy storage models to incorporate multi-year datasets rather than single-year analyses to avoid underestimating storage requirements [27].

Table 1: Common incompatibilities of existing energy storage requirement models.

Incompatibility	Present In	Description
Uses energy generation methods that are not universal to all grids such as nuclear and/or hydropower.	[8], [9], [10], [11], [13], [18]	Nuclear energy is illegal in some countries and hydropower is often constrained by topography and water resources. Consequently, the existing models consider less intermittency in their renewable generation, resulting in lower levels of intermittency and shorter renewable energy drought periods than those observed in high-VRE grids.
Considers a limited number or favours one type of storage technologies.	[9], [10], [11], [13], [14], [17], [18], [19]	PHES and lithium-ion batteries are currently the predominant form of energy storage technologies. However, each region will likely have a preferred type or want to investigate future technologies such as (PtH_2tP) systems. Consequently, the use of these existing models would not allow for practical recommendations for certain electricity grids.
Does not explcitiy model intra-regional transmission constraints or uses single node/copper plate grid representation.	[8], [11], [13], [14], [15], [16], [17]	Aggregated network representations neglect intra-regional congestion and locational constraints, which can significantly underestimate storage requirements and overstate system flexibility. This limitation is particularly relevant for geographically large or weakly interconnected grids.

The definitions of renewable droughts vary considerably in the literature, and some studies define wind droughts as periods when the mean daily wind speed falls into the 25$^{\text{th}}$ percentile of historical data [6], while others focus on weekly averages in the first percentile [28] or periods in which renewable sources operate at 10% or less of the rated capacity [29]. This inconsistency in definition complicates comparative analysis and emphasises the need for standardised approaches.

Gilmore et al. [30], Staffell and Pfenninger [31], and Staffell and Green [32] have investigated the influence of weather events and climate patterns on renewable energy production by developing VRE output forecast models. The development of VRE output forecast models means that decades of historical weather data can be converted into historical wind and solar farm energy output. This allows for the expansion of the VRE datasets that are historically constrained by the relatively recent development and deployment of renewables. The expanded VRE datasets are examined to determine the impact of renewable droughts on the expected energy output and to quantify the risk that renewable droughts occur. For example, Gilmore et al. [30] developed a VRE output forecast model that used 42 years of historical weather data for all current renewable energy locations within the NEM. The results showed that for the NEM operating with 100% renewable energy, '*the worst two-week historical renewable drought on record would cause only a 30% reduction in expected energy output*' [24]. Consequently, Gilmore et al. determined that the worst-case firming capacity necessary for the NEM was the combined total of one-third of the average monthly energy demand and two-thirds of the average daily energy demand [30]. Therefore, the approach of using multiple historical reference years, as employed by Gilmore et al. [30], was adopted in the framework, along with block bootstrapping, to analyse the impact of renewable droughts on the energy storage requirements.

1.1.3. Energy Storage Technology Categorisation

Energy storage systems are typically categorised by technology type and temporal duration. Commercially deployed technologies include grid-scale battery systems (predominantly lithium-ion), PHES, and emerging PtH_2tP systems [7]. Other emerging storage technologies, such as flow batteries, thermal storage, and compressed air energy storage, have recently been trialled within electricity grids around the world

[33]. However, the commercial maturity and grid-scale viability of these technologies remain under active investigation.

Guney and Tepe [34], Shan et al. [35], and Koohi-Fayegh and Rosen [36] provide comprehensive reviews of commercially deployed energy storage systems globally. These reviews highlight several key facts about the current state of energy storage systems: lithium-ion battery systems are currently the systems of choice for short-duration energy storage, while PHES remains the most widely deployed system for long-duration storage due to its cost-effectiveness. PHES can only be used in certain topographies that have sufficiently large differences in elevations, greater than 100 m is preferred, and at least a 3% slope between the top and bottom reservoirs [37]. This often leads to a restriction on the amount of PHES that can be scaled in certain regions.

PtH_2tP produces green hydrogen, by electrolysis, during periods of excess renewable generation, which is stored and used to power hydrogen-fuelled power generators during periods of low renewable generation [38]. Theoretically, hydrogen storage offers the potential for long duration and seasonal storage [38]. However, these benefits have not yet been seen on a grid scale, as PtH_2tP has only recently begun to be built across the world [38]. Such findings underscore the need for a low cost seasonal storage solution that is viable in all electricity grids, regardless of topographical constraints.

Storage categorised according to duration is another area of research that has attracted significant attention, as electricity grids with a high penetration of VRE sources require a wide range of discharge times to overcome intermittency [39]. As a result, energy storage is temporally defined in several ways. In [39], storage was divided into four main categories, very-short-duration storage ($<$ 5 mins), short-duration storage (5 mins to 4 hours), medium-duration storage (4 to 200 hours) and long-duration storage ($>$ 200 hours). However, such a large time range for medium-duration storage is impractical, as this range includes four-hour batteries and PHES, which play significantly different roles in supporting an electricity grid. In [40], storage was split into three categories: short-duration storage ($<$ 10 hours), long-duration storage (10 to 100 hours), and seasonal storage ($>$ 100 hours). However, defining short-duration storage as less than 10 hours covers 93% of the world's energy storage capacity in 2021 [41]. This means that there is a lack of granularity for the short-duration storage definition, leading to energy storage systems with differing roles being classified under the same definition.

In contrast, most studies that examine the temporal energy storage requirements for NEM classify storage according to the temporal categorisations set by AEMO. AEMO defines their storage temporally into three types, shallow storage ($<$ 4 hours), medium storage (4 to 12 hours), and long-term storage ($>$ 12 hours) [7]. Therefore, the case study in this paper uses the temporal storage definitions established by AEMO to specify the quantities of shallow, medium, and deep storage needed in SA.

1.2. Research Gaps and Contributions

The literature review identifies several limitations in the existing analysis of energy storage requirements for RE100 power systems. While a broad range of modelling approaches has been applied, important gaps remain in how spatial dynamics, climatic variability, and system constraints are jointly represented. Critical research gaps include:

1. **Framework Flexibility**: Limited research provides frameworks that are sufficiently flexible to address the unique combination of renewable resource availability and geographic constraints. Many studies abstract lower level grid details through large area aggregation or copper plate representation, limiting their ability to capture nodal congestion and spatial dynamics.
2. **Weather Pattern Integration**: Although the influence of weather on VRE generation is widely acknowledged, most studies rely on single-year datasets or a small number of representative periods for analysis in their modelling. Systematic assessment of storage adequacy across multiple historical weather years remains relatively limited.
3. **Integrated Modelling of System Constraints**: Existing approaches often focus on individual dimensions of the storage sizing problem, such as the focus on a single storage technology, spatial network detail, or temporal cost representation, in isolation. There remains a gap in transparent

frameworks that integrate these dimensions in a manner suitable for long-term storage adequacy assessment for high VRE power systems.

This study addresses the key gaps in long term storage adequacy modelling by developing a generalisable multi-nodal optimisation framework and applying it to the demonstrative case of SA's transition to 100% renewable electricity. The contribution of this work lies within the structured integration of spatial, temporal, and climate dimensions relevant to storage planning in high VRE systems. The key contributions of this work are:

1. **Flexible Multi-Nodal Optimisation Framework**: Development of a comprehensive and flexible multi-nodal DC optimal power flow (OPF) model with linearised transmission loss approximation explicitly designed for long term VRE storage adequacy assessments. The framework is readily transferable to other electricity systems and advances beyond copper-plate or lossless network approximations commonly used in long-horizon studies.
2. **Weather Variability Impact Assessment**: Quantification of storage requirement variations across multiple reference weather years, demonstrating the importance of planning for extreme scenarios rather than for average or single year conditions.
3. **Temporal Storage Classification Integration**: Use of storage classifications (shallow, medium, and deep) with technology specific cost structures and operational constraints, providing practical guidance for storage technology selection and deployment.
4. **Comprehensive Sensitivity Framework**: Systematic evaluation of critical parameters including dispatchable generation integration, reference weather years, and co-optimising storage and generation, providing transparent bounds on storage requirements and policy relevant insights.
5. **Regional Storage Distribution Insights**: Detailed characterisation of nodal-level storage allocation, identifying critical nodes and seasonal patterns essential for infrastructure planning in geographically constrained power systems.

The remainder of this paper is organised as follows. Section 2 presents the methodology outlining the formulation of the multi-nodal grid optimisation problem. Section 3 describes the processes used to apply the developed model to the South Australian electricity grid. Inputs, assumptions, and limitations of the modelled grid are also discussed in this section with particular emphasis on the sources of the input data. Section 4 presents the results of the base case analysis, the regional storage distribution, and the comprehensive sensitivity studies. Section 5 concludes with key findings and policy implications. Detailed data preparation methodologies, optimisation model implementation, and extended sensitivity analyses are provided in the appendices. Furthermore, all code used in this paper is available under an open-source license at `https://github.com/Baz-Payne/MILES-SA`.

2. Multi-Nodal Grid Model

AC OPF formulations are inherently nonlinear and are, therefore, commonly approximated using DC OPF models under a set of simplifying assumptions. The first section of this chapter presents the derivation of a modified DC OPF adopted in this study, while the second section provides a comprehensive discussion of the model limitations introduced by these assumptions and provides their justification.

2.1. Optimisation Formulation

The storage sizing problem is formulated as a mixed-integer linear program. A breakdown of the software and hardware used in the modelling is detailed in Appendix B. The objective function minimises the total present value of storage investment and operating costs across all local nodes r, storage technologies b, and temporal classifications h. C_E and C_P denote the costs associated with the energy and power component of storage, respectively. C_O denotes the present value of operating expenditure, which is assumed to be incurred annually and is proportional to the installed power capacity. Variable operating costs are assumed to be negligible compared to the fixed operating costs for storage technologies, and thus operating costs are modelled as a function of installed capacity rather than dispatch.

$$\min \left(\sum_{r\in\mathcal{R}} \sum_{b\in\mathcal{B}} \sum_{h\in\mathcal{H}} \left(C_{E,b,h} E^{\max}_{r,b,h} + (C_{P,b,h} + C_{O,b,h}) P^{\max}_{r,b,h} \right) \right), \tag{1}$$

where $C_{O,b,h}$ is defined as follows.

$$C_{O,b,h} = C_{A,O,b,h}\text{PVAF}, \quad \forall b \in \mathcal{B}, \forall h \in \mathcal{H} \tag{2}$$

$C_{A,O,b,h}$ denotes the annual operating expenditure for storage type b and temporal classification h. The present value annuity factor (PVAF) is defined as follows.

$$\text{PVAF} = \frac{1 - (1 + dsc)^{-N}}{dsc}, \tag{3}$$

where dsc is the capital discount rate and N is the analysis period.

Energy Balance Constraint

The energy balance constraint is defined in (4), illustrating that at any time t, the supply and demand at node r must be balanced. Unlike most contemporary models, nodal generation is a parameter in this model. This is reflective of the overall aim of the model, to minimise storage, and the availability of comprehensive renewable energy planning models. To capture increasing consumer behavioural shifts during periods of reliability triggers or high spot market prices, demand-side participation (DSP) is incorporated, stipulating that entities can provide additional non-scheduled generation or reduce their load. Although restricted to a small amount in many electrical grids, energy at node r can go unserved which is captured by the USE term. Finally, the transmission line losses are explicitly modelled.

$$G_{r,t} + P_{\text{disc},r,t} + P_{I,r,t} - P_{\text{chrg},r,t} - P_{\text{curt},r,t} + P_{\text{DSP},r,t} - D_{r,t} + \text{USE}_{r,t} - P_{loss,r,t} = 0 \quad \forall r \in \mathcal{R}, \forall t \in \mathcal{T}, \tag{4}$$

where $G_{\mathrm{r,t}}$ is the sum of all utility scale generation at node r,

$$G_{r,t} = \sum_{g\in G(r)} G_{r,g,t}, \tag{5}$$

$$G_{r,g,t} \geq 0, \quad \forall r \in \mathcal{R}, \forall t \in \mathcal{T}. \tag{6}$$

where $P_{\text{disc},r,t}$ is the cumulative discharge power at time t for all types and temporal classifications of storage at node r subject to

$$P_{\text{disc},r,t} = \sum_{b\in\mathcal{B}} \sum_{h\in\mathcal{H}} P_{\text{disc},r,b,h,t}, \tag{7}$$

$$0 \leq P_{\text{disc},r,b,h,t} \leq P^{\max}_{r,b,h} \quad \forall r \in \mathcal{R}, \forall b \in \mathcal{B}, \forall h \in \mathcal{H}, \forall t \in \mathcal{T}. \tag{8}$$

$P_{I,r,t}$ is the total power flow into/out of node r at time t for all nodes connected to node r. Power flows are restricted by thermal line limits $P^{\max}_{ij,t}$ of the line ij at time t as depicted in (10).

$$P_{I,r,t} = \sum_{ij \text{ or } ji\in\mathcal{L}(\mathrm{r})} P_{ij,t}, \tag{9}$$

$$|P_{ij,t}| \leq P^{\max}_{ij,t} \quad \forall t \in \mathcal{T}. \tag{10}$$

$P_{\text{chrg},r,t}$ is the cumulative charge power at time t for all types and temporal classifications of storage at node r subject to,

$$P_{\text{chrg},r,t} = \sum_{b\in\mathcal{B}} \sum_{h\in\mathcal{H}} P_{\text{chrg},r,b,h,t}, \tag{11}$$

$$0 \le P_{\text{chrg},r,b,h,t} \le P_{r,b}^{\max} \quad \forall r \in \mathcal{R}, \forall b \in \mathcal{B}, \forall h \in \mathcal{H}, \forall t \in \mathcal{T}. \tag{12}$$

$P_{\text{curt},r,t}$ is the curtailment at time t at node r, constrained by (13).

$$P_{\text{curt},r,t} \ge 0 \quad \forall r \in \mathcal{R}, \forall t \in \mathcal{T}. \tag{13}$$

The dispatchable amount of DSP, $P_{\text{DSP},r,t}$, is limited by the maximum DSP capacity, $P_{\text{DSP}}^{\max}$, as shown in (14). Furthermore, the daily nodal DSP amount is limited to hrs_{DSP} hours of equivalent operation. Let d denote the current day in the year such that $d = \{1, \cdots, 365\}$ or during a leap year, $d = \{1, \cdots, 366\}$. Therefore, the daily nodal DSP constraint is given by (15). Lastly, the annual DSP is limited by an amount $E_{\text{DSP}}^{\max}$ as stipulated in (16).

$$0 \le P_{\text{DSP},r,t} \le P_{\text{DSP}}^{\max} \quad \forall r \in \mathcal{R}, \forall t \in \mathcal{T}. \tag{14}$$

$$\sum_{r\in\mathcal{R}} \sum_{t=\frac{24(d-1)}{\Delta t}}^{\frac{24d}{\Delta t}-1} P_{\text{DSP},r,t}\Delta t \le hrs_{\text{DSP}} P_{\text{DSP}}^{\max} \quad \forall d. \tag{15}$$

$$\sum_{r\in\mathcal{R}} \sum_{t\in\mathcal{T}} P_{\text{DSP},r,t}\Delta t \le E_{\text{DSP}}^{\max}. \tag{16}$$

The instantaneous power USE is limited by $\text{USE}^{\max}$. Furthermore, the annual amount of USE within all local nodes is limited to a percentage $\text{USE}_{\%}$ of the total forecast demand within all local nodes, as shown in (18).

$$\text{USE}_{r,t} \le \text{USE}^{\max} \quad \forall r \in \mathcal{R}, \forall t \in \mathcal{T} \tag{17}$$

$$\sum_{r\in\mathcal{R}} \sum_{t\in\mathcal{T}} \text{USE}_{r,t}\Delta t \le \text{USE}_{\%} \sum_{r\in\mathcal{R}} \sum_{t\in\mathcal{T}} D_{r,t}\Delta t \tag{18}$$

Transmission line losses are approximated using the first order linearisation of the quadratic I^2R losses around a reference operating point, inspired by works [42] and [43]. The loss on line ij at time t is given by

$$P_{ij,t}^{loss} = R_{ij} P_{ij,t}^2 \tag{19}$$

where R_{ij} denotes the line resistance and $P_{ij,t}$ denotes the flow on the line. Linearising (19) around a reference flow $\bar{P}_{ij,t}$ results in (20)

$$P_{ij,t}^{loss} \approx R_{ij}{\bar{P}_{ij,t}}^2 + 2R_{ij}\bar{P}_{ij,t}(P_{ij,t} - \bar{P}_{ij,t}) \tag{20}$$

Constant terms are omitted because they do not affect optimal solutions. To ensure non-negative losses under bidirectional power flows, the absolute value of line flow is considered, resulting in the following linear approximation.

$$P_{ij,t}^{loss} \approx 2R_{ij}\bar{P}_{ij,t}|P_{ij,t}| \tag{21}$$

The reference flow is taken as 50% of the transmission line rating at time t, $0.5P_{ij,t}^{\max}$. Thus, $\beta_{ij,t}$ can be defined as (22), representing the marginal increase in losses per unit of line flow.

$$\beta_{ij,t} = 2R_{ij}(0.5P_{ij,t}^{\max}) = R_{ij}P_{ij,t}^{\max} \tag{22}$$

The transmission line losses are assumed to be split evenly at either end of the line, and hence half the losses are incurred at each node. Therefore, losses at node n can be defined as (23) and implemented using a piecewise linear formulation to preserve model linearity, as shown in (24). At the optimal point, only one of $P_{pos,ij,t}$ or $P_{neg,ij,t}$ will be non-zero due to their contribution to losses.

$$P_{loss,r,t} = \sum_{ij \text{ or } ji \in \mathcal{L}(\mathrm{r})} 0.5\beta_{ij,t}|P_{ij,t}| \quad \forall r \in \mathcal{R}, t \in \mathcal{T} \tag{23}$$

$$P_{loss,r,t} = \sum_{ij \text{ or } ji \in \mathcal{L}(\mathrm{r})} 0.5\beta_{ij,t}(P_{pos,ij,t} + P_{neg,ij,t}) \quad \forall r \in \mathcal{R}, t \in \mathcal{T} \tag{24}$$

$$P_{ij,t} = P_{pos,ij,t} - P_{neg,ij,t} \quad \forall ij \in \mathcal{L}, t \in \mathcal{T} \tag{25}$$

$$P_{pos,ij,t} \geq 0 \quad \forall ij \in \mathcal{L}, t \in \mathcal{T} \tag{26}$$

$$P_{neg,ij,t} \geq 0 \quad \forall ij \in \mathcal{L}, t \in \mathcal{T} \tag{27}$$

DC Power Flow

Active power flow between two nodes is determined by the difference in the voltage angles at each node as given in (28).

$$P_{ij,t} = B_{ij}(\theta_{i,t} - \theta_{j,t}) \quad \forall ij \in \mathcal{L}, \forall t \in \mathcal{T}, \tag{28}$$

where $B_{i,j}$ is the matrix of the reciprocals of the line reactance between nodes i and j. The reference voltage angle constraint is defined as follows.

$$\theta_{ref,t} = 0, \quad \forall t \in \mathcal{T}. \tag{29}$$

Storage System Dynamics

The energy of storage at node r, of storage type b, of temporal classification h, at time $t+1$, $E_{r,b,h,t+1}$, is determined by the amount of charge $P_{\mathrm{chrg},r,b,h,t}$ and discharge $P_{\mathrm{disc},r,b,h,t}$ and the energy stored $E_{r,b,h,t}$ in the previous time step, t. This constraint is depicted in (30).

$$E_{r,b,h,t+1} = E_{r,b,h,t} + \left(\eta_{c,b}P_{\mathrm{chrg},r,b,h,t} - \frac{P_{\mathrm{disc},r,b,h,t}}{\eta_{d,b}}\right)\Delta t \quad \forall r \in \mathcal{R}, \forall b \in \mathcal{B}, \forall h \in \mathcal{H}, \forall t \in \mathcal{T}, \tag{30}$$

where $\eta_{c,b}$ is the charging efficiency of the storage type b, $\eta_{d,b}$ is the discharge efficiency of the storage type b, and Δt is the sampling interval. To ensure that the charging at node r ceases once the storage is at its maximum capacity $E^{\max}_{r,b,h}$ and to ensure that the storage is not discharged beyond its minimum capacity $E^{\min}_{r,b,h}$, (31) is used. The former constraint, in conjunction with the supply and demand balance equation, ensures that any additional energy is curtailed, and the latter ensures that the storage is kept above its minimum level.

$$E^{\min}_{r,b,h} \leq E_{r,b,h,t} \leq E^{\max}_{r,b,h} \quad \forall r \in \mathcal{R}, \forall b \in \mathcal{B}, \forall h \in \mathcal{H}, \forall t \in \mathcal{T} \tag{31}$$

Each storage type is temporally limited by the definitions outlined previously. The classifications are shallow storage, $<$ 4 hours, medium storage, 4 to 12 hours, and deep storage, $>$ 12 hours. These constraints are expressed by the following.

$$4P^{\max}_{r,b,shlw} \leq E^{\max}_{r,b,shlw} \quad \forall r \in \mathcal{R}, \forall b \in \mathcal{B} \tag{32}$$

$$4P^{\max}_{r,b,med} \geq E^{\max}_{r,b,med} \quad \forall r \in \mathcal{R}, \forall b \in \mathcal{B} \tag{33}$$

$$12P^{\max}_{r,b,med} \leq E^{\max}_{r,b,med} \quad \forall r \in \mathcal{R}, \forall b \in \mathcal{B} \tag{34}$$

$$12P^{\max}_{r,b,deep} \geq E^{\max}_{r,b,deep} \quad \forall r \in \mathcal{R}, \forall b \in \mathcal{B} \tag{35}$$

To ensure that the storage does not simultaneously discharge and charge, the big-M constraint method is used, shown below.

$$P_{\mathrm{disc},r,b,h,t} \leq \alpha_{r,b,h,t} M \quad \forall r \in \mathcal{R}, \forall b \in \mathcal{B}, \forall h \in \mathcal{H}, \forall t \in \mathcal{T} \tag{36}$$

$$P_{\mathrm{chrg},r,b,h,t} \leq (1 - \alpha_{r,b,h,t}) M \quad \forall r \in \mathcal{R}, \forall b \in \mathcal{B}, \forall h \in \mathcal{H}, \forall t \in \mathcal{T}, \tag{37}$$

$$\alpha_{r,b,h,t} \in \{0, 1\}. \tag{38}$$

M is chosen as a large number that exceeds the largest expected maximum power requirement but not so large that it causes computational errors.

Storage Specific Constraints

Depending on the storage technology used in the optimisation model, additional technology specific constraints may be required. The constraints presented below are not intended to be exhaustive, but are used to illustrate how the model captures key operational and physical limitations associated with different storage technologies.

Battery technologies experience capacity degradation throughout their service life, which is strongly correlated with cumulative cycling. As such, batteries are typically subject to a limited total number of cycles before replacement is required. To prevent excessive degradation, the optimisation model imposes a limit on the number of cycles that can be completed over a time frame. The constraint below limits the number of cycles, N_{cycles}, over the time period for battery storage technologies $\mathcal{B}_{\mathrm{bat}}$.

$$\sum_{t \in \mathcal{T}} P_{disc,b,h,t} \Delta t \leq N_{cycles} E^{\max}_{r,b,h} \quad \forall r \in \mathcal{R}, \forall b \in \mathcal{B}_{\mathrm{bat}}, \forall h \in \mathcal{H} \tag{39}$$

The deployable capacity of PHES at a given location is constrained by geographic factors such as reservoir availability and elevation differences. Thus, it is essential that the optimisation model correctly encapsulates these site specific limitations. The following defines the build limits for different temporal classifications, h, of PHES in terms of reservoir availability and elevation differences, respectively.

$$E^{\max}_{r,\mathrm{PHES},h} \leq E^{lim}_{r,\mathrm{PHES},h} \quad \forall r \in \mathcal{R}, \forall h \in \mathcal{H} \tag{40}$$

$$P^{\max}_{r,\mathrm{PHES},h} \leq P^{lim}_{r,\mathrm{PHES},h} \quad \forall r \in \mathcal{R}, \forall h \in \mathcal{H} \tag{41}$$

Neighbouring Nodes Energy Balance

The final constraint is provided in the instance that the full modelling of neighbouring grids is infeasible. Note that this constraint is not necessary to the functioning of the previous constraints. The simplified energy balance constraint on neighbouring nodes is defined in (42).

$$P_{I,s,t} - P_{\mathrm{curt},s,t} - P_{loss,s,t} + GS_{s,t} = 0 \quad \forall s \in \mathcal{S}, \forall t \in \mathcal{T}, \tag{42}$$

where $GS_{s,t} \geq 0$ if and only if there is surplus generation in the neighbouring node s.

2.2. Model Limitations

The simplifying assumptions adopted in the optimisation model formulation introduce limitations in the representation of real-world electricity systems. Although these assumptions constrain the scope of analysis, they also highlight areas for future model improvement. This section discusses the main limitations of the modelling framework and provides justification for the assumptions adopted.

The exclusion of a specific economic dispatch layer is one of the more significant limitations of the derived optimisation model. Conventional power system planning models typically combine a capacity outlook model and a time sequential dispatch model. In contrast, the formulation of the optimisation model in this paper focuses on the determination of storage sizing. The absence of an explicit dispatch layer is partially mitigated through the inclusion of a constraint that limits excessive storage cycling, discouraging unrealistic operating behaviour. Furthermore, in an electricity grid with VRE sources, wind and solar generators are not dispatched solely on the basis of spot market prices, as their operation is influenced by contractual agreements, subsidies, and other market incentives, providing additional justification for this modelling choice.

Another limitation of the model is the assumption that storage proponents will participate exclusively in the electricity spot market. In practice, storage proponents will participate in a range of markets to generate revenue streams, including the spot market [44], energy and frequency control markets [45], contract markets, and other bilateral services to recover costs and generate a profit. Consequently, not all storage available in the model will be available to balance electricity supply and demand. However, this assumption reduces the computational complexity of the problem that would otherwise need to be solved with multiple market categories while still allowing the model to address the overarching question of determining storage adequacy.

While not explicitly stated, the model places a greater emphasis on the transmission network compared to the distribution network. This reduced granularity of the dispatch network may underestimate the congestion and operational constraints that arise at the distribution level. Furthermore, the potential contribution of distributed energy resources, such as residential battery systems, is not explicitly modelled. While the aggregate capacity of distributed storage may become comparable to that of utility-scale assets in the future, this study adopts a utility scale perspective, with this potential captured in the overarching nodal storage requirements.

A further limitation is the assumption that the modelled electricity grid can operate without synchronous generation. While the requirement of synchronous generators is declining, most electricity grids still maintain a non-zero synchronous inertia requirement. For example, a minimum of 42 megawatts (MW) of synchronous generators is required in SA. Consequently, the modelled electricity grid would likely need synchronous condensers or grid forming inverters to support system strength in a high VRE grid, thus increasing the overall cost. However, as the capital cost of energy storage dominates the overall system cost, the exclusion of these synchronous technologies is not expected to materially affect the study's conclusion regarding storage capacity requirements.

3. South Australia: A Case Study

SA exemplifies the renewable energy transition, having increased renewable energy generation from 1% to more than 70% during the past two decades [21]. This rapid transformation positions SA as a world leader in the integration of renewable energy within a gigawatt-scale grid. The State Government [46] and ElectraNet [21] have announced that SA will achieve net 100% renewable energy generation by financial year 2027-28. This target would place SA among the first gigawatt-scale electricity systems globally to reach this milestone, making it an ideal case study for evaluating the proposed model. However, compared to many of the other high-renewable energy electricity grids, SA faces more challenging operating conditions due to its sparse network, limited interconnection, and high dependence on VRE sources.

To determine the storage requirements for a RE100 South Australian electricity grid, it is first crucial to describe the context in which the problem is being solved. This section begins with a brief overview of AEMO and their forecasting framework. The methodology used to create generation and demand traces from AEMO's forecasting data is also discussed. The second part describes the model inputs and their respective data sources. Finally, the assumptions underpinning the case study are described along with justifications. A visual representation of the modelling workflow is presented in Figure 2.

3.1. Forecasting Body

To plan for the future requirements of the NEM, AEMO publishes an integrated system plan (ISP) every two years [7]. AEMO's method of determining future developments is based on a complex linear program,

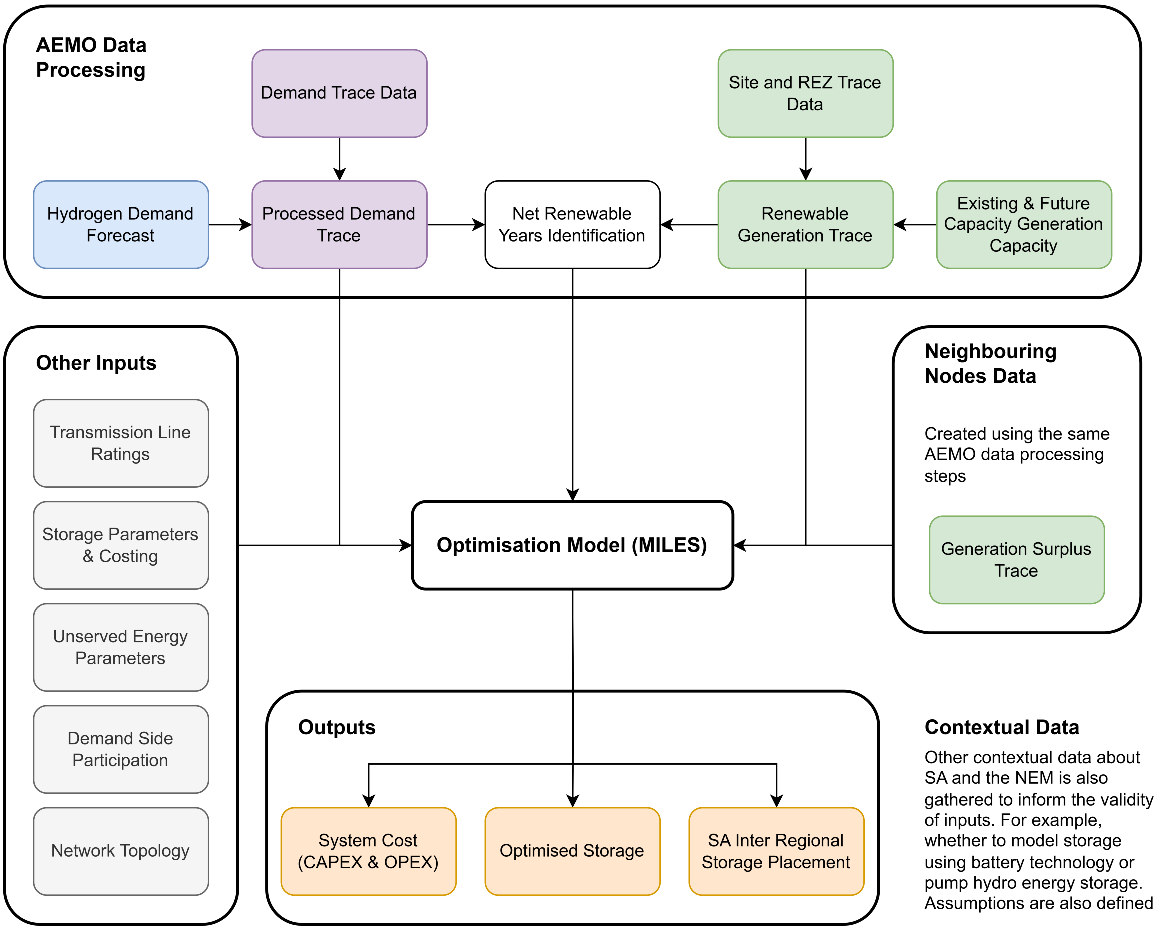

Figure 2: Data processing workflow and model inputs.

which incorporates a range of inputs and assumptions to model more than 1,000 simulations that takes multiple days to complete [47]. The model produces an Optimal Development Path (ODP) that optimises the benefits to consumers given the uncertainties in the future outlook [48].

Three future scenarios are presented to encapsulate different rates of future development, from slow to rapid advancement: *Progressive Change*, *Step Change*, and *Green Energy Exports*, with AEMO deeming *Step Change* as the most likely scenario. For the *Step Change* scenario, the ODP is Candidate Development Path (CDP) 14. Thus, all forecasts and figures used in the modelling of this study are extracted from AEMO's 2024 ISP Step Change scenario CDP14. The following segments outline how AEMO's data is used to create the trace data used in the modelling.

3.1.1. Generation Forecast

AEMO defines renewable energy zones (REZs) for each NEM state which are areas of potential renewable energy development based on 10 development criteria. Refer to Appendix C.1 for more information. Furthermore, AEMO also provides half-hourly generation data for solar and wind for each existing generator and each REZ within the NEM. Existing generator data is based on wind and solar data taken from the site. Data for REZs is determined using mesoscale wind flow modelling at turbine hub height for wind modelling, and global horizontal irradiance and direct normal irradiance data from satellite imagery for solar modelling [49]. Furthermore, AEMO uses 13 years of historic data in the forecast modelling to account for short- and medium-term weather diversity [49]. Each trace is given as floating-point numbers between 0 and 1, with 0 representing no generation across a half-hour interval and 1 representing the maximum output within an existing site or REZ.

To create generation traces, the generation trace data can be combined with AEMO's forecasted solar and wind generation capacities. The historic and forecast aggregated solar and wind generation capacities for SA, taken from AEMO's 2024 ISP generation and storage outlook spreadsheet [50], are shown in Figure 3. During the ISP's forecast period, from FY2024-25 to FY2051-52, the capacity of the generation sites and REZs in SA is expected to change. Existing generators are approaching the end of their useful life, as shown by the decreases in solar and wind capacities in the 2030s. In contrast, REZs are expected to have increased capacity as new generators are constructed. In FY2051-52, SA is projected to have almost four times the installed renewable generation capacity it has today. Furthermore, the expected split between solar and wind is forecast to be approximately even, compared to the present conditions where there is significantly more wind installed.

3.1.2. Demand Forecast

AEMO uses various terms to describe what is occurring in the NEM. A key forecast term is demand, which '*describes the amount of electricity needed at a particular time in units of watts'* [51]. AEMO uses several demand terms; however, the one of interest for this paper is operational demand, which refers to electricity used by residential, commercial and large industrial consumers, after the impact of photovoltaic nonscheduled generation, other nonscheduled generation, rooftop PV and nonvirtual power plant battery storage systems. Thus, when the term "demand" is used throughout the paper, it refers to operational demand.

Similar to generation, AEMO provides half-hour demand traces for the forecast period. One caveat is that these demand traces exclude hydrogen due to the slow technical progress and unproven commercial viability of hydrogen, which makes it an uncertainty [47]. However, SA is observing interest in large hydrogen proponents, some of the order of GW [52], which would substantially alter the forecast demand. Thus, hydrogen demand was manually incorporated into the demand traces using AEMO's latest forecasts [53].

When adding the hydrogen load to the demand, a few assumptions were made. First, it was assumed that the hydrogen load is flexible and could be added to any desired half-hour period. This assumption is justified through studies conducted on the ramp rates of electrolysers, e.g., Deloitte [54] and Macquarie [55], which demonstrate ramp rates are in the range of 10% of the total capacity each second. The second assumption is that hydrogen plants will consume electricity when there is net negative demand in the system, producing "green hydrogen". This assumption is two fold as renewable generators may not generate during periods of low demand due to spot market pricing and hydrogen producers may have signed agreements to

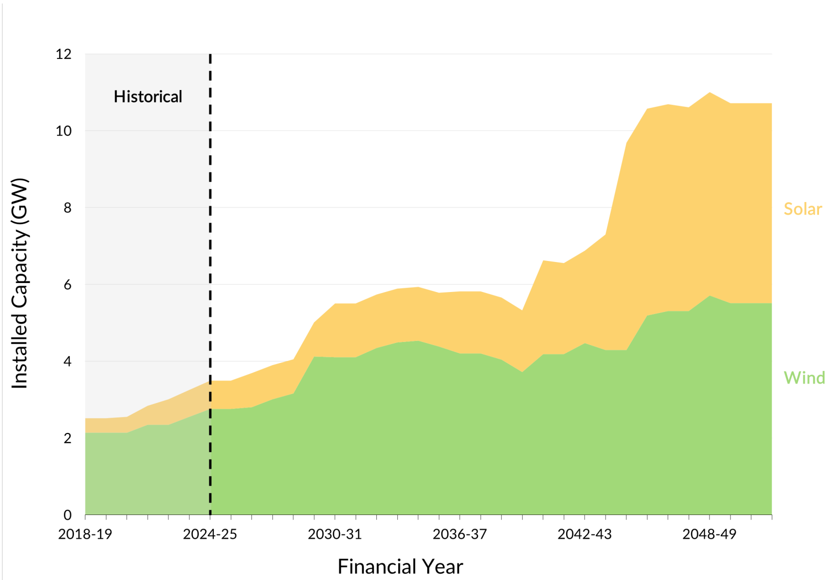

Figure 3: Forecast and historic solar and wind generation capacity in SA.

produce a certain amount of hydrogen each month. If there are limited net negative demand times, then hydrogen plant owners will be unable to reach their quota. However, due to the uncertainty surrounding the operation of these plants, these assumptions are considered reasonable.

Another demand element is DSP, which refers to activities taken by entities in an electricity grid to reduce demand [56]. This can be done by reducing a non-scheduled load or supplying unscheduled generation [56]. AEMO considers DSP as a dispatchable capacity, which means that it can be used alongside renewable energy generation to reduce demand. Furthermore, SA Power Networks has recently launched flexible connections, helping to lower peak network demand and simultaneously rewarding customers with lower tariffs [57]. One caveat to this dispatchable capacity is that AEMO limits the DSP to a maximum 2-hour continuous operation each day [58].

3.1.3. Identification of Net Renewable Years

To identify net renewable years for the case study, renewable generation and demand traces were aggregated over each calendar year (CY) and combined, as illustrated in Figure 4. The data is presented in CYs as opposed to FYs for several reasons. Firstly, the start of a CY coincides with the middle of summer in Australia, where renewable generation is typically the highest, meaning that any assumptions around starting state of charge are justified. Secondly, demand and generation remain relatively constant in net zero years, as shown in Figure 4, which means that there are few changes between FYs. As such, switching the analysis to CYs does not result in a loss of generality. Based on this aggregation, the period 2045 to 2051 is selected as the primary case study, representing seven years of historic weather variability. This period is chosen as renewable generation is consistently larger than demand. Each year is modelled using its corresponding demand and renewable generation traces, with no explicit coupling between years or probabilistic scenario sampling across years. Multi-year climate variability in the base case is therefore represented through the use of multiple historic weather years rather than through stochastic formulations or cross year coupling. However, additional weather years are incorporated through a sensitivity analysis to more extensively depict climate variability.

The importance of considering multiple weather years is illustrated in Figure 5 and Figure 6. These figures present the weekly average solar and wind generation in SA, expressed in per unit of the maximum generation capacity. Three years from the forecast period are shown, corresponding to the best, worst and average cases based on weekly solar and wind generation. Notably, the years selected for both solar

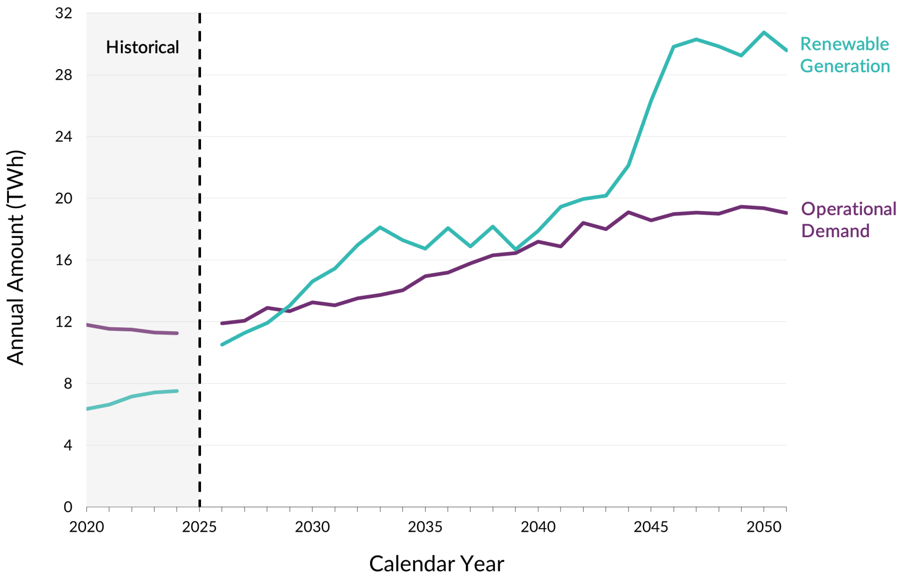

Figure 4: Historical and forecast annual operational demand and renewable generation based on AEMO's 2024 ISP and latest hydrogen forecasts. Note that the current calendar year is not included as the data has not yet been released by AEMO.

and wind coincided, however, this alignment occurred by chance. Both solar and wind generation exhibit seasonal variations: solar generation increases in the summer weeks while wind generation tends to decrease, highlighting the complementary nature of these renewable sources.

Similar to the case of generation, three years from the forecast period are shown in Figure 7 that illustrate the weekly average demand in per unit of the FY2024-25 average operational demand in SA ∼1300 MW [21].

3.2. Model Inputs

Many nodal divisions of SA are possible, but the one used in this project is a mix of the ElectraNet regional division and the AEMO REZs. The nodal division is shown in Figure 8. A more detailed figure that outlines the network configuration is presented in Appendix C.2. The nodal division was chosen for two reasons: first, to remain aligned with the one used by ElectraNet's planning team and second, to ensure that the AEMO REZs could be allocated to each node.

A limitation of the chosen nodal decomposition is that AEMO's demand trace is provided for SA as a whole rather than individual nodes. Consequently, demand is divided into each node according to data provided by ElectraNet. Generation data is similarly spatially allocated by mapping AEMO REZs to the model nodes, as illustrated in Figure 8. More details on the demand division and REZ to node mapping are provided in Appendix C.3.

The multi-nodal grid incorporates constraints based on summer and winter transmission line ratings which are published by ElectraNet [59]. Transmission capacity is treated as an exogenous input to the optimisation problem rather than being an optimisation parameter. This ensures consistency with the objective of determining the storage requirement of an electricity grid rather than determining the optimal transmission expansion. Dynamic line ratings were not considered due to the limited availability of complete and temporally consistent weather datasets across the South Australian network. Individual line ratings are extrapolated and aggregated to determine inter nodal transfer capacities, as detailed in Appendix C.4. Anticipated future transmission growth is represented by explicitly including planned network expansion projects. For inter nodal branches where no committed future upgrades are currently identified, the transfer capacity is assumed to increase by 10% relative to current limits. This assumption reflects conservative

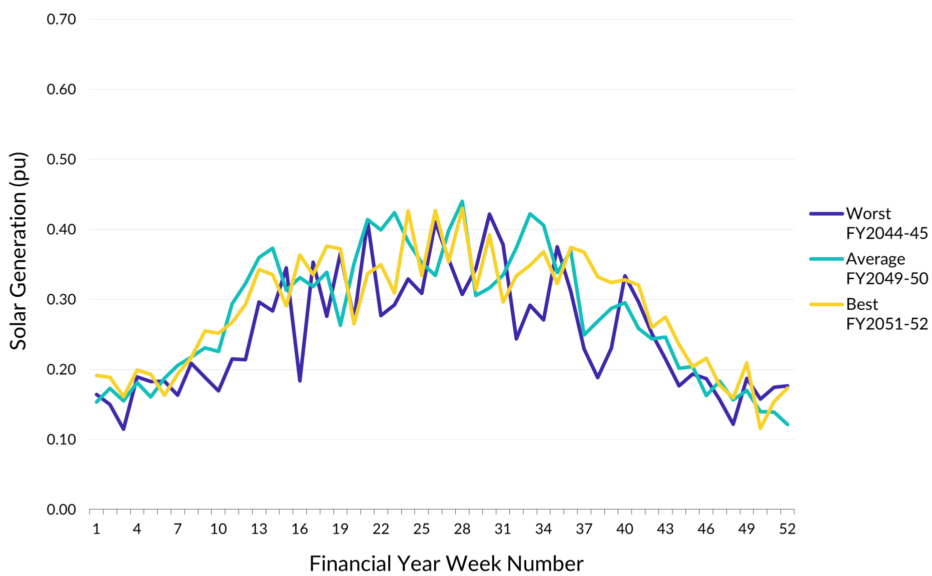

Figure 5: Comparison of the weekly average solar generation across the best, worst and representative years within the forecast period.

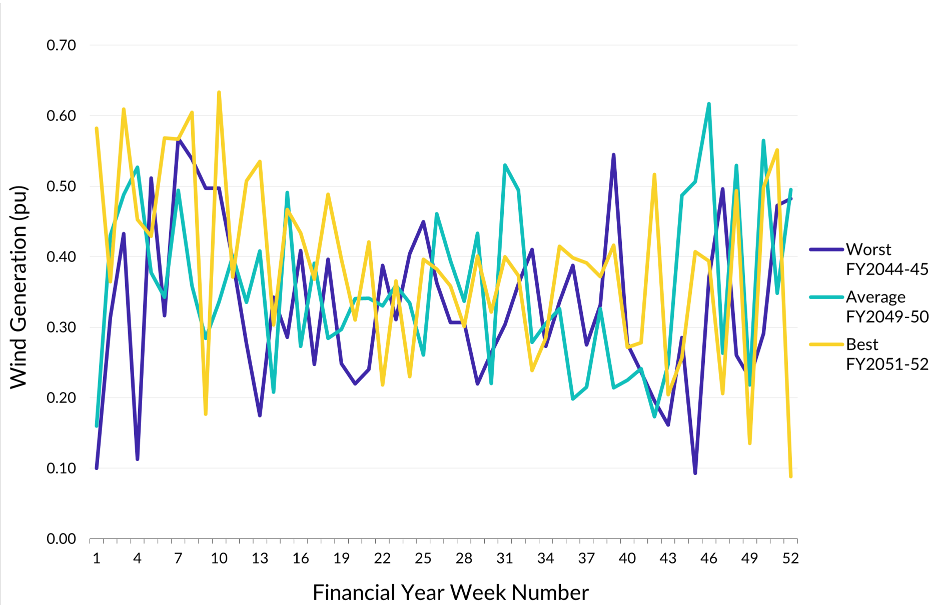

Figure 6: Comparison of the weekly average wind generation across the best, worst and representative years within the forecast period.

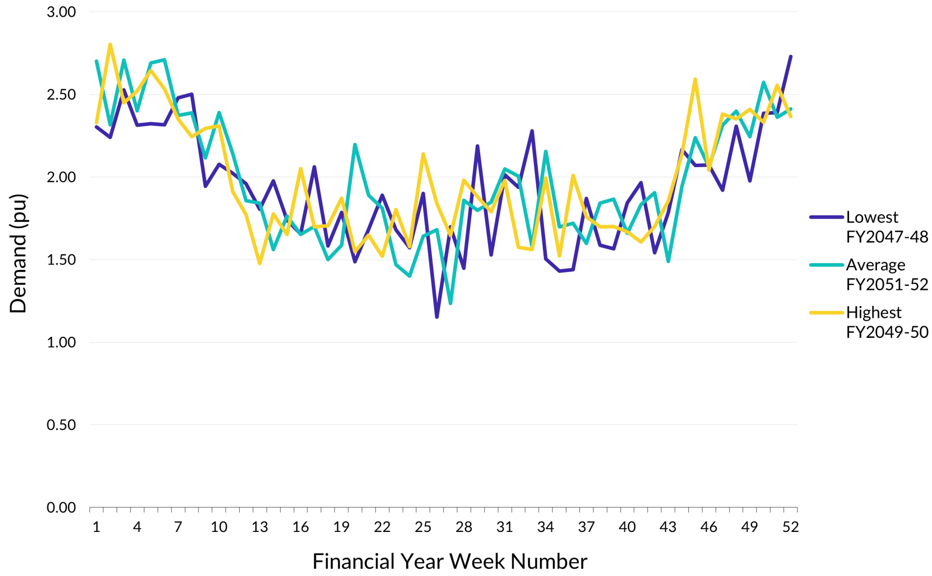

Figure 7: Comparison of the weekly average demand across the best, worst and representative years within the forecast period. Demand is in per unit of the FY2024-25 average operational demand in SA (1,300 MW).

incremental network augmentation over the modelling horizon while ensuring that the optimisation prioritises efficient use of the existing transmission infrastructure. Appendix C.4 also outlines the process used to estimate transmission line parameters including resistance and reactance.

Future cost projections are inherently uncertain, and forecasts are subject to ongoing scrutiny. Previous editions of CISRO's *GenCost* reports have been controversial due to their inclusions and exclusions of certain costs in their levelised cost of energy [60]. However, the costs used in the model are related to the power and capacity component of the storage rather than the levelised cost and therefore, the GenCost pricing is used in the model.

All remaining input parameters and their sources are summarised in Table 2. Monetary values are expressed in real $AUD2025–26 terms unless stated otherwise. All storage cost parameters include balance of plant costs. For computational purposes, values are converted to per-unit equivalents in the model backend, but are presented in absolute terms for clarity.

A limitation of the adopted temporal classifications is that each category spans a range of storage durations. For example, the shallow storage category includes systems with durations from 0 to less than 4 hours. As storage capacity costs generally decline with increasing system size due to economies of scale, representative costs are required for each category. Accordingly, the median/representative duration within each temporal classification is used to determine storage cost parameters.

Finally, PHES is spatially constrained to the Iron Triangle node, consistent with the REZ definition described in Appendix C.1.

Table 2: Detailed list of model input parameters and their source.

Parameter	**Value**	**Unit**	**Description**
S_{base}	10	GVA	System base chosen for solver speed.
Shallow	<4	hours	As defined by AEMO [61].
Medium	4-12	hours	As defined by AEMO [61].

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Table 2 continued from previous page

Parameter	Value	Unit	Description
Deep	>12	hours	As defined by AEMO [61].
$C_{E,\text{LIB,shlw}}$	366,000	\$ /megawatt hour (MWh)	Cost taken from GenCost 2 hour battery storage for the year 2045 [62].
$C_{E,\text{LIB,med}}$	209,000	\$ /MWh	Cost taken from GenCost 8 hour battery storage for the year 2045 [62].
$C_{E,\text{LIB,deep}}$	179,000	\$ /MWh	Cost taken from GenCost 24 hour battery storage for the year 2045 [62].
$C_{P,\text{LIB,shlw}}$	732,000	\$ /MW	Cost taken from GenCost 2 hour battery storage for the year 2045 [62].
$C_{P,\text{LIB,med}}$	1,672,000	\$ /MW	Cost taken from GenCost 8 hour battery storage for the year 2045 [62].
$C_{P,\text{LIB,deep}}$	4,296,000	\$ /MW	Cost taken from GenCost 24 hour battery storage for the year 2045 [62].
$C_{A,O,LIB,shlw}$	13,392	\$ /MW/year	Based on a linear interpolation of AEMO's 2025 ISAR fixed O&M costs for a 2 hour battery [63].
$C_{A,O,LIB,med}$	38,001	\$ /MW/year	Based on a linear interpolation of AEMO's 2025 ISAR fixed O&M costs for an 8 hour battery [63].
$C_{A,O,LIB,deep}$	103,625	\$ /MW/year	Based on a linear interpolation of AEMO's 2025 ISAR fixed O&M costs for a 24 hour battery [63].
$C_{E,PHES,med}$	339,000	$\$/MWh$	Cost taken from GenCost 10 hour PHES for the year 2045 [62].
$C_{E,PHES,deep}$	139,000	$\$/MWh$	Cost taken from GenCost average of 24 and 48 hour PHES for the year 2045 [62].
$C_{P,PHES,med}$	3,389,000	$\$/MWh$	Cost taken from GenCost 10 hour PHES for the year 2045 [62].
$C_{P,PHES,deep}$	4,580,000	$\$/MWh$	Cost taken from GenCost average of 24 and 48 hour PHES for the year 2045 [62].
$C_{A,O,PHES,med}$	96,740	$\$/MW/year$	Based on a linear interpolation of AEMO's 2025 ISAR fixed operating and maintenance costs for a 10 hour battery [63]

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Table 2 continued from previous page

Parameter	**Value**	**Unit**	**Description**
$C_{A,O,PHES,deep}$	80,195	$\$/MW/year$	Based on the average of AEMO's 2025 ISAR fixed operating and maintenance costs for a 24 hour battery [63]
dsc	3.90%	%	Discount rate as defined in [64].
N	20	years	Current technical life of grid scale batteries as defined by AEMO [63]. N can be >20 years, however, replacement costs of certain storage technologies would need to be considered.
PVAF	13.71		Calculated from the above parameters
$\frac{\eta_{c,LIB,shlw}}{\eta_{d,LIB,shlw}}$	92/92 (RTE = 84.6)	%	Based on AEMO's 2025 ISAR storage properties for a 2 hour battery
$\frac{\eta_{c,\text{LIB, med}}}{\eta_{d,\text{LIB, med}}}$	92.5/92.5 (RTE = 85.6)	%	Based on AEMO's 2025 ISAR storage properties for an 8 hour battery
$\frac{\eta_{c,LIB,deep}}{\eta_{d,LIB,deep}}$	93/93 (RTE = 86.5)	%	Based on AEMO's 2025 ISAR storage properties for a 24 hour battery
$\frac{\eta_{c,PHES,med}}{\eta_{d,PHES,med}}$	87.2/87.2 (RTE = 76)	%	Based on AEMO's 2025 ISAR storage properties for PHES
$\frac{\eta_{c,PHES,deep}}{\eta_{d,PHES,deep}}$	87.2/87.2 (RTE = 76)	%	Based on AEMO's 2025 ISAR storage properties for PHES
$P_{ij,t}^{\max}$	Refer Appendix C.4	MW	Based on ElectraNet's summer and winter transmission line ratings
X_{ij}	Refer Appendix C.4	pu	Estimated using typical /km values and publicly available line length data
R_{ij}	Refer Appendix C.4	pu	Estimated using typical /km values and publicly available line length data
Δt	0.5	hours	Extrapolated from AEMO's ISP workbook [65].
$P_{DSP}^{\max}$	Refer Appendix C.5	MW	Extrapolated from AEMO's ISP workbook [65].
$E_{DSP}^{\max}$	Refer Appendix C.5	MWh	Extrapolated from AEMO's ISP workbook
hrs_{DSP}	2	hours	
$\text{USE}^{\max}$	300	MW	Corresponds to ∼10% of the maximum demand for SA [7].

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Table 2 continued from previous page

Parameter	**Value**	**Unit**	**Description**
$\text{USE}_{\%}$	0.002%		As defined in the Rules [66].
$\beta_{ij,t}$	N/A	%	Calculated as defined in (22).
M	100,000		Chosen sufficiently large to ensure discharge and charging do not occur simultaneously while not causing computational issues.
N_{cycles}	1	cycles/day	Chosen to be aligned with current industry guidelines.
$E^{lim}_{\text{Iron Triangle},PHES,med}$	5,400	MWh	As outlined in [67].
$E^{lim}_{\text{Iron Triangle},PHES,deep}$	62,400	MWh	As outlined in [67].
$P^{lim}_{\text{Iron Triangle},PHES,med}$	100	MW	As outlined in [67].
$P^{lim}_{\text{Iron Triangle},PHES,deep}$	1800	MW	As outlined in [67].

One of the initial approaches taken to solve the multi-nodal grid problem was to allow the optimisation model to determine the flows of the interconnectors between SA and connected NEM States. However, many REZs of SA, NSW and Victoria are not geographically distant from one another, meaning that if renewable energy generation is low in SA, there is a high probability it could be low in NSW and Victoria. As an additional investigation, the correlation between the energy generation of SA, NSW and Victorian REZs was investigated, inspired by the work carried out in [68]. It is noted that correlation, as a statistical measure of weather, has limitations [69] and is only accurate to the first degree, but for illustrative purposes, it is used here. Presented in Figure 9 is the relationship between the correlation of REZs generation and their geographical distance, measured from the approximate centre point between REZs, for all solar and wind REZs in SA, NSW and Victoria. As depicted in the figure, the correlation of generation between REZs is related to distance, highlighting the importance of interconnectors to provide access to a geographically diverse range of generators. The relationship between solar REZs and distance is not as strong as wind REZs whose generation trace correlation is closely related to the separation of sites. However, in general, solar REZs are more correlated than wind REZs.

Therefore, to create a more accurate representation of the interconnector reliance, demand and generation traces were created for NSW and Victoria to determine the amount of renewable energy generation available for import to SA. To reduce model complexity, NSW and Victoria are considered as single node grids, with all demand and generation aggregated at a single point. It is acknowledged that this assumption abstracts significant nuances of these grids including the intra nodal network constraints and operation characteristics.

Only solar and wind renewable energy traces are considered, with hydro power explicitly excluded. This decision was made due to the complexity involved in developing accurate hydro traces for NSW and Victoria given the relatively limited knowledge about these electricity grids. Excluding hydropower also results in a more conservative estimate of import availability, thereby avoiding an overstatement of interconnector support to SA and providing a clear avenue for future model extension.

3.3. Assumptions

As in any research, outlining the assumptions used in modelling is crucial. These assumptions help differentiate the research in this paper from others, justify specific decisions, and suggest directions for future studies. Detailed below are the specific assumptions made in this model formulation, their limitations and justification.

Figure 8: Regional mapping of SA and visualisations of the connections between regions.

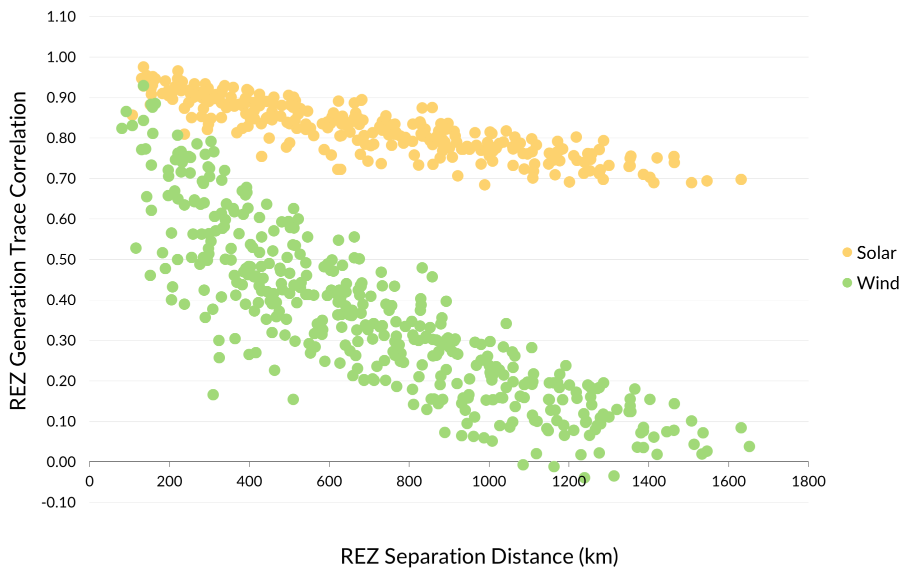

Figure 9: Relationship between the generation trace correlation and distance of solar and wind REZs in SA, NSW, and Victoria.

One of the key assumptions of the modelling is the accuracy of AEMO's forecast data. Both future demand and generation are uncertain as demonstrated in the vast differences between the *Step Change* and *Green Energy Exports* scenarios. Furthermore, renewable generation is strongly influenced by weather variability, meaning that forecast generation traces represent approximations of future conditions. Generation capacities and traces could differ in the future and aspects, such as climate change, could significantly affect generation. Despite these uncertainties, no alternative datasets or forecasting methodologies of comparable scope, transparency, and public availability currently exist. To mitigate the limitations of this assumption, two sensitivity analyses are conducted. The first incorporates block bootstrapping of historical weather years to model the importance of considering multiple weather years. The second derives an alternative model formulation in which renewable generation capacity is co-optimised with storage rather than represented as a fixed value.

Another assumption related to generation siting is that all development will occur in AEMO's defined REZs. While developments outside these zones may occur, particularly for emerging generation and storage technologies, the REZs reflect AEMO's comprehensive planning process and extensive stakeholder consultation. As such, this assumption provides a consistent and policy aligned basis for spatial allocation of future generation while acknowledging that deviations could occur.

One of the most significant assumptions made is the exclusion of dispatchable generation. As electricity grids with intermittent sources continue to strive for 100%, the amount of dispatchable generation will decrease. However, it is likely that there will be small amounts of dispatchable generation, renewable or non-renewable, installed in such networks. The exclusion of dispatchable generation therefore represents a simplifying assumption that may overstate storage requirements. To address the limitations of this assumption, a targeted sensitivity analysis is conducted.

Finally, the model represents SA's transmission network (132kV, 275kV and 330kV) and excludes the lower distribution voltages. As previously discussed, this omits details and constraints that may be imposed at both a transmission and distribution level. However, developing such a model is infeasible due to the consultation and inputs required from SA Power Networks and ElectraNet. Furthermore, the limited computing resources available for this paper would not likely be able to solve such an optimisation problem in a reasonable timeframe. Given these practical limitations, the adopted network model is a trade-off between model accuracy and computational practicality.

Having outlined the inputs and assumptions of the model, the results are presented in the following section.

4. Results and Discussion

The first section below outlines the base case results of the model. Key observations are identified, which then motivate a series of sensitivity analyses that test the inputs and assumptions of the underlying model. Note that for conciseness, the results presented in the following sections are only a summary of the model outputs. More detailed results are provided in the relevant appendices, while complete results are accessible through the accompanying GitHub repository available at `https://github.com/Baz-Payne/MILES-SA`. The section concludes with a comparison of the storage requirement found in this study with similar studies on the South Australian storage requirement.

4.1. Base Case Storage Requirements

The optimisation problem is solved for each CY from 2045 to 2051 and the results are shown in Table 3. In the base case, it is assumed that SA can import from other states only when surplus wind and solar generation is available interstate. Under these conservative assumptions, it can be seen that the collective energy storage requirement for SA was 7.278 GW/644.2 GWh in 2051, with an estimated system cost of \$139 billion. See Appendix D for the results in detail. Notably, winter consistently emerges as the critical period for storage sizing across all years, reflecting seasonal patterns of low renewable generation coinciding with increased heating demand.

Despite storage requirements peaking at 2051, there is no sudden decline in renewable energy generation compared to other years. See Appendix D for further detail. For example, 2049 has less total renewable energy generation and a higher demand but a storage need of over 300 GWh less. While 2051 is a seemingly extreme case, the total renewable energy generation exceeds demand by 55%, representing a plausible future scenario that grid planners may need to be prepared for.

Table 3: Multi-nodal grid base case energy storage requirements by year for SA.

Year	Power (GW)	Energy (GWh)	System Cost ($B)	Critical Period
2045	5.084	418.8	92.71	Winter
2046	4.248	198.4	49.08	Winter
2047	5.099	275.3	62.93	Winter
2048	4.699	242.6	60.25	Winter
2049	5.774	307.4	72.25	Winter
2050	4.895	203.2	50.69	Winter
2051	**7.278**	**644.2**	**139.2**	Winter
Median	5.084	275.3	62.93	-

The regional and temporal split of storage for 2051 is outlined in Table 4. In particular, most of the storage was allocated to the Adelaide Metro node, which is the largest load centre in the state. However, the 644.2 GWh of energy storage is an infeasible amount, as this amount would be sufficient to power SA for around two thirds of a month assuming an average operational demand of ∼1300 MW [21]. This capacity is almost double the size and is estimated to cost approximately 7 times that of the Snowy 2.0 project [70]. Compared to the studies discussed previously, the required 644.2 GWh exceeds the SA requirements and is, instead, more closely aligned with the overall NEM requirements. To address these issues, several sensitivity analyses are conducted in the following section.

4.2. Sensitivity Analysis

To investigate the validity of the assumptions and to explore additional cases of interest, several sensitivities are investigated in the following sub-sections. The first sensitivity analysis investigates storage

Table 4: Regional storage split for the worst year, 2051, in the investigation period.

Node	Shallow		Medium		Deep	
	Power (GW)	Energy (GWh)	Power (GW)	Energy (GWh)	Power (GW)	Energy (GWh)
Battery Storage						
Adelaide Metro 275	0	0	1.802	21.62	1.486	469.5
Eyre Peninsula 132	0.0512	0.2048	0.0877	1.052	0.0391	15.05
Iron Triangle 275	0.2224	0.8895	0.8954	9.626	0	0
Upper North 275	0.0586	0.2344	0.3222	2.289	0	0
Mid North 330	0	0	0	0	0.3017	13.83
Riverland 132	0	0	0.5212	6.255	0.4885	6.225
South East 275	0	0	0.7314	8.777	0.1258	17.36
Pumped Hydro Energy Storage						
Iron Triangle 275	N/A	N/A	0	0	0.1453	62.40
Total	**0.3322**	**1.329**	**4.360**	**49.62**	**2.587**	**593.3**

reduction by incorporating a dispatchable generation unit to supplement renewable resources during winter droughts. The second sensitivity analysis explores multiple weather patterns and employs block bootstrapping to test the model's robustness. The final sensitivity analysis tests AEMO's forecast installed renewable energy capacity by co-optimising storage and generation, as is commonly seen in the literature.

4.2.1. Dispatchable Energy Unit

One of the key themes present throughout this study has been the issue of the intermittent nature of renewable energy. The two renewable energy sources investigated in SA are not dispatchable, increasing the required storage, as the generation cannot be dispatched when required. As predicted by AEMO [7], SA is likely to use other forms of generation in the future to ensure that supply and demand can be balanced. Thus, it is unlikely and infeasible for SA to develop large amounts of storage to support 100% renewable operation, made of solar and wind, when it is more economical to have small amounts of other dispatchable energy sources. To test this hypothesis, a generic dispatchable generator is added to SA's largest demand node, Adelaide Metro 275, as depicted in the updated supply and demand constraint shown below.

$$\begin{gathered} G_{r,t} + P_{\text{disc},r,t} + P_{I,r,t} - P_{\text{chrg},r,t} - P_{\text{curt},r,t} + P_{\text{DSP},r,t} - D_{r,t} + \text{USE}_{r,t} - P_{loss,r,t} = 0, \\ r = \text{ Adelaide Metro 275}, \forall t \in \mathcal{T} \end{gathered} \tag{43}$$

The annual dispatchable generation constraint is shown in (44). Here, DISP% denotes the percentage of South Australian demand that the dispatchable generator can serve.

$$\sum_{t \in \mathcal{T}} P_{\text{disp},t} \Delta t \leq \text{DISP\%} \sum_{t \in \mathcal{T}} \sum_{r \in \mathcal{R}} D_{r,t} \Delta t \tag{44}$$

To keep this sensitivity analysis open ended in nature, the type of dispatchable energy used and, hence cost, are not specified, meaning that the generation could be from gas, hydrogen, biomass, or a mix of these technologies. Different generator capacities from 200 to 1,000 MW are added to the Adelaide Metro 275 node for this analysis. These units are assumed to have fast ramp rates. However, the annual amount of generation is limited to 1% to 3% of SA's total demand or ∼200 GWh to ∼600 GWh. For reference, SA used ∼600 GWh of gas generation in the month of June alone in 2024 [71]. The results for the worst-case year 2051 are shown in Figure 10. Detailed results for each year are provided in Appendix E.

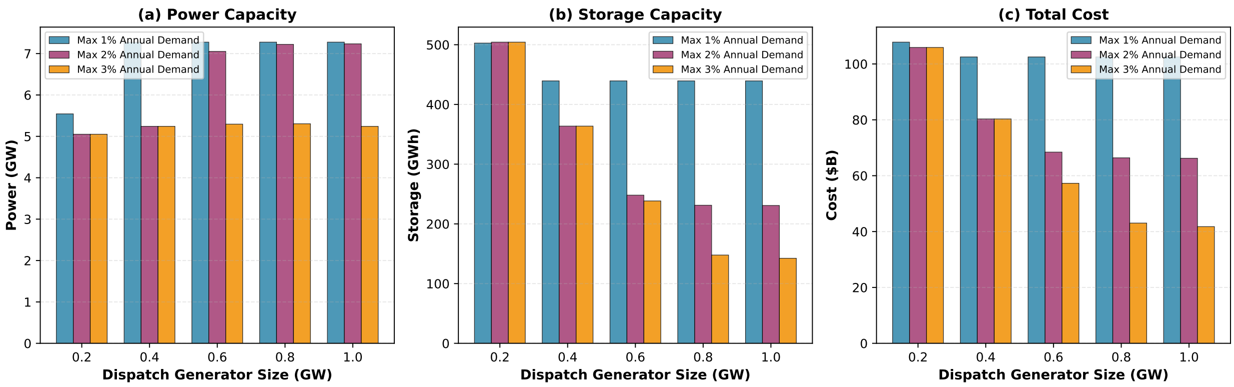

Figure 10: Summary of the worst-case storage requirements for different generator dispatch sizes and percentages of demand met.

Meeting just 1% of the total demand in SA with dispatchable generators results in a storage requirement of 439.4 GWh, a 32% reduction in overall storage, and an approximate $35 billion reduction in cost. Despite increases in generator capacity, storage is limited to a minimum of 439.4 GWh for capacities greater than or equal to 400 MW. This result indicates that the generator reaches saturation when supplying only 1% of demand; further storage reductions require covering a larger share of total demand. When 2% of the total demand is served by the dispatchable generator, the storage requirement is reduced to 230.8 GWh. As the requirements did not saturate between 800 MW and 1,000 MW, there is potential for the storage requirement to be reduced further at 2% demand for larger generator capacities. A similar result is seen when 3% of the demand is met with the storage requirement being reduced by 78% or to 142 GWh. This analysis highlights significant storage savings when relatively small amounts of dispatchable generation are used. Such forms of generation will be crucial, especially at times of low generation and high demand.

An issue with this sensitivity analysis is the potential scenario it creates for exercises of market power. Since there is only a short window in which additional generation is required, owners of dispatchable generation units can exercise market power, increasing costs for consumers. Therefore, it is imperative that deterrents, such as price caps, be implemented to protect consumers in these scenarios.

4.2.2. Multiple Weather Years

While the base case considered seven weather years, the literature clearly articulates the importance of assessing a larger number of years to thoroughly model multi-year climate variability and the uncertainty of VRE resources. To explore this variability, two approaches were taken.

The first approach evaluates the storage requirement for the worst-case year 2051 under AEMO's complete set of 13 reference weather years. The second approach extends the analysis by bootstrapping AEMO's reference weather data to generate synthetic reference years. This enables further exploration of possible climate variability beyond AEMO's limited historical record. In each case, the 2051 demand trace is fixed and the generation profiles vary according to the weather year.

13 Years of Weather

The results for the 13 reference weather years are shown in Table 5. The significant range of storage requirements highlights the sensitivity of storage needs to annual climate variations. This also reinforces the importance of considering multiple weather years to account for the uncertainty of VRE generation.

A key finding from this sensitivity analysis is the importance of the temporal distribution of renewable energy generation. As described in Appendix F, the difference in total renewable energy generation in SA for the reference year proceeding the worst-case year is 63.87 GWh ($\sim$0.2% of the total amount of renewable generation). Despite this relatively minor difference in renewable generation, the storage differs by almost 400 GWh. This finding reiterates the importance of not only the quantity of renewable energy, but its timing relative to demand.

Table 5: Storage requirement statistics for 13 years of reference data.

Parameter	Minimum	Maximum	Range	Average
Power (GW)	4.238	6.030	1.792	5.085
Energy (GWh)	159.3	677.9	518.5	340.6
System Cost ($B)	43.45	141.6	98.13	77.40
Renewable Energy Generation (GWh)	2,803	3,156	354	3,002

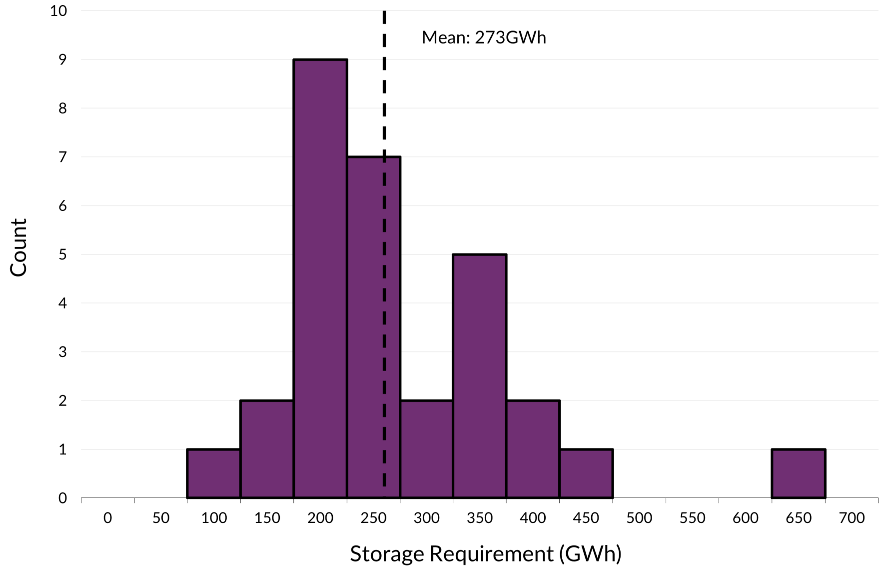

Figure 11: Histogram illustrating the storage requirements for the 30 block bootstrap weather years with a bin size is 50GWh.

Synthetic Reference Years

30 years of synthetic wind and solar traces were generated using a month-conditioned multivariate block bootstrap applied to the AEMO's trace data. Wind and solar outputs at all nodes were resampled jointly to preserve correlations between VRE technologies and across spatial dimensions. Blocks of seven days were sampled with replacement, with block start times restricted to the same calendar month in order to preserve seasonal weather patterns. Synthetic years were constructed by stitching and trimming blocks to standard monthly durations. This approach preserves diurnal structure, synoptic-scale persistence, and wind–solar dependence while avoiding seasonal inaccuracies.

Figure 11 illustrates the storage requirements obtained from the 30 years of synthetic weather. Compared to the reference weather dataset, the mean storage requirements across the synthetic years is approximately 70 GWh lower. As observed in the reference weather dataset, the bootstrap analysis highlights the annual climate variability of the storage requirements. However, the majority of the synthetic years group around a similar amount, with a relatively low number of years resulting in significantly higher or lower amounts of storage.

4.2.3. Co-Optimising Generation and Storage

A significant assumption made throughout this work is the accuracy of the AEMO's forecast generation data. This sensitivity analysis investigates the potential storage savings when generation is co-optimised with storage. This is also a common practice in the literature on SA's storage requirement for a RE100 grid to co-optimise both generation and storage.

The optimisation objective is modified as follows where $\mathcal{G}$ denotes the set of generation technologies considered (solar and wind), C_P denotes the costs associated with the power component, and C_O denotes the present value of the operating expenditure. The previously defined energy balance constraint is extended to incorporate both existing generator sites and potential new generator sites. The amount of generation that can be developed at each site is also constrained by technology specific build limits. The complete optimisation reformulation with a discussion on the adopted build limits is provided in Appendix G.1.

$$\min\left(\sum_{r\in\mathcal{R}}\left(\sum_{b\in\mathcal{B}}\sum_{h\in\mathcal{H}}\left(C_{E,b,h}E^{\max}_{r,b,h}+(C_{P,b,h}+C_{O,b,h})P^{\max}_{r,b,h}\right)\right)+\sum_{g\in\mathcal{G}}(C_{P,g}+C_{O,g})P^{\max}_{VRE,r,g}\right) \tag{45}$$

The cost parameters for solar and wind are reported in Table 6.

Table 6: Generation cost parameters.

Technology	**Cost of Power** ($\$/MW$)	**Operating Expenditure** ($\$/MW/year$)
Large Scale Solar	896,000 [62]	12,220 [63]
Wind	2,324,000 [62]	28,510 [63]

The results of the co-optimisation problem are given in Table 7, with detailed results presented in Appendix G.2. Co-optimising storage and generation results in an approximate 460 GWh reduction in storage requirement and a $70 billion reduction in the overall system cost, noting that the base case assumes zero cost for installed renewable capacity. The installed solar and wind capacities are ∼30% higher than AEMO's or, in absolute terms, approximately 3GW. The results highlight that oversizing the VRE generation capacity relative to demand is a lower cost strategy to achieve a 100% renewable grid.

Another notable shift in storage requirements emerges when storage and generation are co-optimised. Compared to the base case where 2051 had the largest storage requirements and cost, the co-optimised results display more diverse outcomes across years. Year 2045 now has the largest storage requirement, while the system cost is less than the results from 2051. This outcome reflects the interaction between overcapacity of renewable generation and storage. In 2045, the developed renewable energy capacity is 13.6 GW compared to 2051, which has 17.2 GW, thus increasing the overall system cost. Overall, these results depict the trade-off between oversizing renewable capacity and storage capacity, demonstrating that higher storage requirements do not necessarily correspond to a higher total system cost.

Table 7: Multi-nodal grid co-optimisation energy storage and renewable energy requirements by year for SA.

Year	**Storage Power** (GW)	**Storage Energy** (GWh)	**Storage System Cost** ($B)	**Solar Capacity** (GW)*	**Wind Capacity** (GW)*	**Total System Cost** ($B)
2045	4.587	**180.7**	46.80	7.838	5.768	70.80
2046	4.969	102.9	33.04	9.487	3.978	53.93
2047	4.056	97.0	30.52	6.587	4.671	50.20
2048	4.384	134.0	38.54	8.604	**7.716**	68.64
2049	6.042	83.9	31.26	9.860	7.276	61.50
2050	4.572	112.6	34.52	7.544	4.614	55.07
2051	**6.290**	156.4	**47.47**	**10.59**	6.596	**76.63**
Median	4.587	112.6	34.52	8.604	5.768	61.50

*Not inclusive of the limited number of existing sites that may be operational in each of the analysed future years.

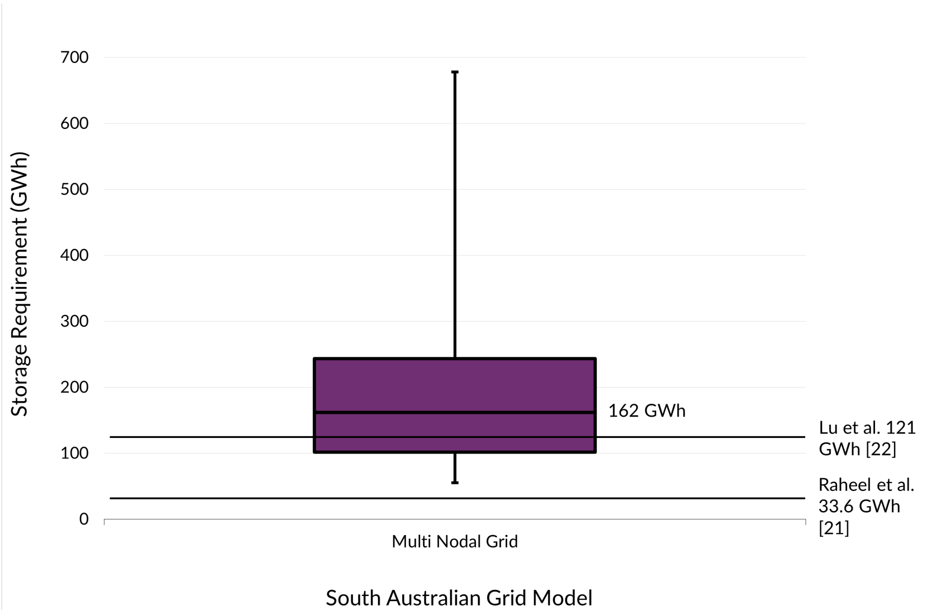

Figure 12: Box and whisker plot comparing the storage requirement of all cases and years considered in the multi nodal grid. Previous research results are also displayed for comparison.

4.3. Model Comparison

A comparison of the results for all years in the investigation period and across all sensitivity analyses is shown in Fig. 12. The research suggests that with current data, the median storage requirement for SA is 162 GWh, with a maximum potential of 678 GWh required in the worst case scenario under conservative operating assumptions. The significant disparity between the worst case and the median storage requirement highlights the trade-off between cost and the achievement of a truly RE100 grid under extreme operating conditions that system planners must consider. This also raises the question of the feasibility of designing a system that is 100% renewable as opposed to operating within a few percent of a 100% renewable grid. From both economic and environmental perspectives, the optimal solution may be somewhere just below 100% renewable generation in the worst years and 100% in most years.

Compared to previous studies by Raheel et al. [16] and Lu et al. [17], the storage requirements in this study are considerably higher. The differences arise from several modelling choices including the consideration of a broader set of scenarios, the inclusion of multiple weather reference years, the granularity of the modelled electricity grid, and power flow constraints. It can be seen that analysing multiple future scenarios is critical due to the inherent uncertainty of future events and developments within electrical grids. Considering multiple weather years draws inspiration from Gilmore et al. [30] and is substantiated by the considerable variations in weather and renewable generation observed in SA in recent years. Representing SA with 13 nodes captures transmission constraints that are often abstracted in previous studies, while modelling power flow explicitly allows the model to consider congestion effects, which further increases storage requirements.

5. Conclusion

This study presents the first comprehensive multi-nodal optimisation analysis of energy storage requirements for South Australia's grid transition to 100% renewable electricity. Key findings include:

1. The worst-case storage requirement, under conservative operating conditions, of 7.28 GW/644 GWh in the base case represents a substantial target, although economic considerations favour hybrid systems

with modest dispatchable generation. Sensitivity analyses demonstrated that this number could be as high as 7.44 GW/678 GWh in the worst case scenario.

2. Annual weather variability results in a substantial range of storage needs, demonstrating the importance of considering multiple weather years.
3. Co-optimising storage and generation results in a lower overall system cost and suggests that oversizing renewable generation capacity relative to demand may be a low cost strategy to achieve a 100% renewable grid.
4. Winter periods consistently drive storage sizing because of the extended low-renewable generation coinciding with higher demand, necessitating long-duration storage solutions.
5. Modest dispatchable generation (1-3% of annual demand) can reduce storage requirements by 32-78%, suggesting that hybrid systems may be the most economic optimal.
6. The geographical correlation of renewable sources limits the effectiveness of interstate support during system-wide drought conditions.

This study advances our understanding of the challenges of the integration of renewable energy and provides practical guidance to grid planners and policymakers. Future work should examine the integration of emerging storage technologies, sub-5 MW home batteries, demand response programs, and sector coupling opportunities to further optimise renewable energy systems.

The methodological framework developed is applicable to other jurisdictions facing similar challenges in the transition to renewable energy, particularly isolated power systems with limited interconnection. As renewable energy costs continue to decline and storage technologies advance, the findings provide a foundation for developing cost-effective and reliable 100% renewable energy systems.

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Declaration of Competing Interest

The authors declare that they have no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The processed data required to reproduce the above findings is available at `https://github.com/Baz-Payne/MILES-SA`.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT and Grammarly to check grammar and improve phrasing. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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Appendix A. Interconnector Outages

Table A.1 shows the list of all interconnector outages in SA from 2010 to the present date. The cause of events is also listed to determine whether there are any common denominators for interconnector outages. As shown in the table, there appear to be two main causes of interconnector outages: weather and equipment/protection malfunction. Weather is the single largest cause of outage time.

The last row displays the total outage time of the Heywood interconnector and the Murraylink interconnector since 2010. Heywood has been down for approximately 588 hours and Murraylink has been down for approximately 412.4 hours. The total time SA has been islanded from the rest of the NEM, both Heywood and Murraylink down, since 2010 is approximately 409 hours. Adjusted per year, this corresponds to $\sim$29 hours per year.

When observing the list of events, the most extreme has occurred within the last 5 years. When just the last 5 years are considered, SA has an average island time from the rest of the NEM of $\sim$82 hours per year. This justifies the necessity of assessing conditions under which the SA grid remains islanded for extended durations.

Table A.1: Major Heywood and Murraylink Interconnector Outage Events

Date	Event Description	Cause	Heywood Downtime	Murraylink Downtime
19 Oct 2011 [72]	Heywood interconnector outage	Malfunction of line protection	35 min	N/A
27 Nov 2012 [73]	Murraylink interconnector outage	Lightning	N/A	106 min
13 Dec 2012 [74]	Heywood interconnector outage	Equipment failure	14 min	N/A
30 Sep 2013 [75]	Murraylink interconnector outage	Storm	N/A	92 min
1 Nov 2015 [76]	Heywood interconnector outage	Incompatible protection relay configuration	35 min	N/A
28 Sep 2016 [77]	Statewide blackout	Weather events and simultaneous tripping of multiple wind farms	65 min	>65 min
1 Dec 2016 [78]	Heywood interconnector outage	Equipment failure	265 min	N/A
25 Aug 2018 [79]	Heywood interconnector outage	Lightning strikes triggering separations	14 min	N/A
16 Nov 2019 [80]	Heywood interconnector outage	Equipment malfunction	292 min	Out of service at the time
31 Jan 2020 [81]	Loss of Heywood interconnector; SA islanded from Victoria	Weather	17 days (∼24,480 min)	17 days (∼24,480 min), Capacity was set to 0 MW
12 Nov 2022 [82]	Loss of Heywood interconnector	Transmission tower failure	7 days (∼10,080 min)	N/A
Total			**∼588 hours**	**∼412.4 hours**

Appendix B. Optimisation Model Implementation

The specifications of the system used to solve the optimisation problem are illustrated below. Solution times range from minutes for single years to hours for sensitivity studies in multiple scenarios.

Hardware

- **Processor (CPU):** AMD Ryzen 7 7735HS 3.2GHz 8 cores 16 threads (chipset for laptop)
- **Memory (RAM):** LPDDR5-6400 32GB
- **Storage:** M.2 2280 SSD 1TB
- **Graphics Card (GPU):** AMD Radeon 680M

Software

- **Operating System:** Windows 11 Home
- **Programming Language:** Python 3.12.3
- **Development Environment:** Spyder 5.5.1
- **Python Package:** PuLP 2.9.0
- **Optimisation Solver:** Gurobi 13.0.0

Appendix C. Supplementary Information for SA Case Study Inputs

The following sub-appendices outline the definition of REZs, the South Australian grid topology, the REZ and demand nodal mapping, and the transmission line parameter formulations, respectively.

Appendix C.1. Renewable Energy Zones

REZs are identified based on 10 development criteria [63] that were originally created in a report by Det Norske Veritas [83]. The report details how each criterion was mapped, weighted, and scaled for the assessment of REZs. The development criteria, as outlined by AEMO, are reproduced below:

1. Wind resource - a measure of high wind speeds (above 6m/s).
2. Solar resource – a measure of high solar irradiation (above 1,600 kW/m2).
3. Demand matching – the degree to which local resources correlate with demand.
4. Electrical network – the distance to the nearest transmission line.
5. Cadastral parcel density – an estimate of the average property size.
6. Land cover – a measure of the vegetation, water bodies, and urbanisation of areas.
7. Roads – the distance to the nearest road.
8. Terrain complexity – a measure of terrain slope.
9. Population density – the population within the area.
10. Protected areas – exclusion areas where development is restricted.

The original REZs identified by Det Norske Veritas in conjunction with the consultation of the 2018, 2020, 2022, and 2024 ISPs have led to the REZs as seen in the current draft of the 2026 ISP [63]. Thus, it should be noted that in future iterations of the ISP, the REZs may change. The current REZs in the NEM are depicted in Figure C.1. In particular, SA has potential for some PHES development in S5 Northern SA, which corresponds to the Iron Triangle node.

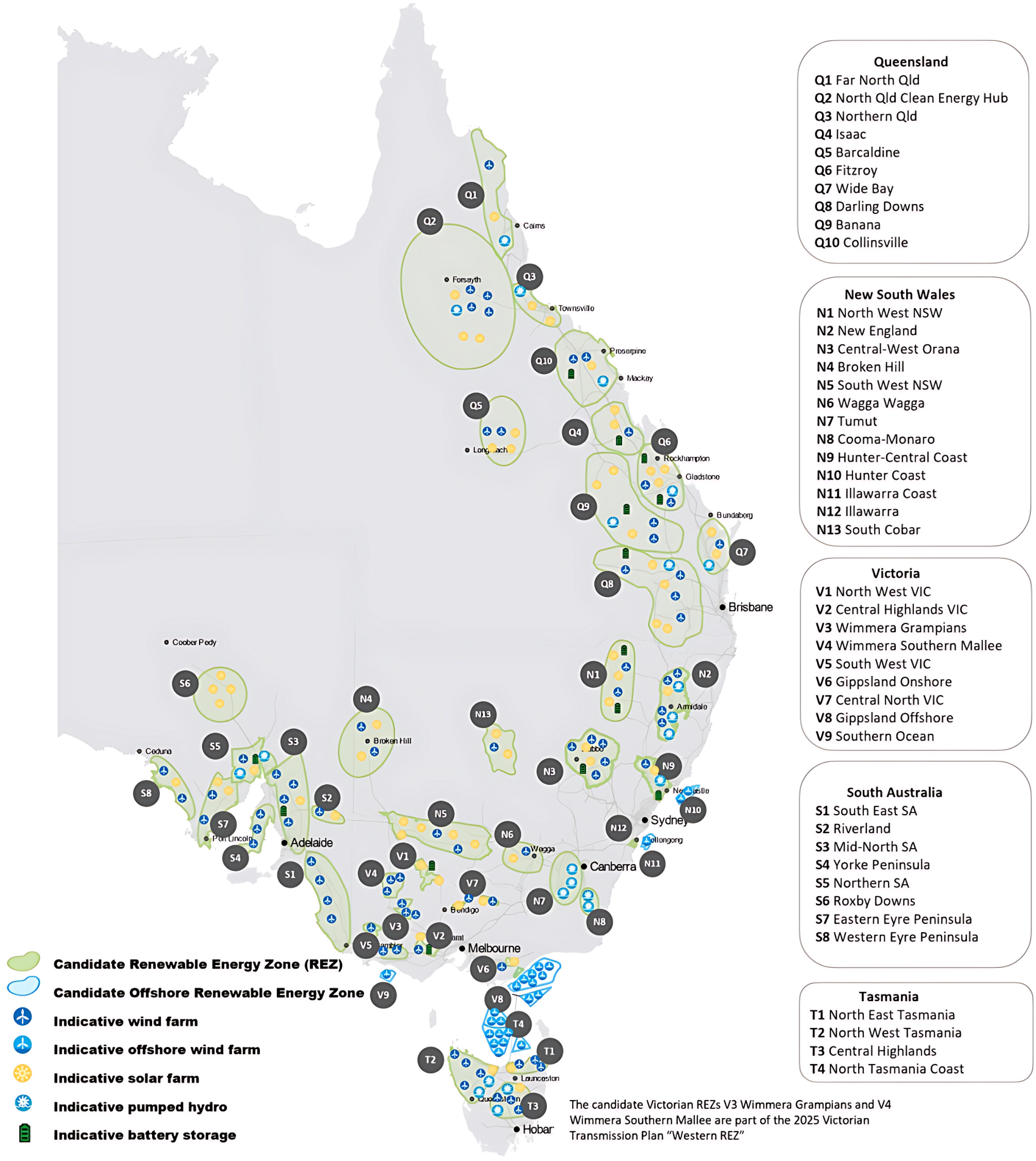

Figure C.1: AEMO's REZs as of the draft 2026 ISP [84].

Appendix C.2. South Australian Grid Topology

The South Australian electricity grid is modelled by nodes, and branches that connect these nodes as illustrated in Figure C.1. Load and generation are assumed to occur on the high voltage bus of each region for simplification as this study focuses on storage sizing rather than power flows.

Appendix C.3. REZ and Demand Nodal Mapping

AEMO's renewable energy zones correspond to the nodal division of SA as defined in Table C.1. A similar process is repeated for all current generators in the state.

Table C.1: AEMO REZ mapping to SA nodal division.

AEMO REZ	**SA Grid Node**
S1 South East SA	South East 275
S2 Riverland	Riverland 132
S3 Mid-North SA	Mid North 330
S4 Yorke Peninsula	Mid North 330
S5 Northern SA	Iron Triangle 275
S6 Roxby Downs	Upper North 275
S7 Eastern Eyre Peninsula	Eyre Peninsula 132
S8 Western Eyre Peninsula	Eyre Peninsula 132

The demand in SA is mapped according to Table C.2.

Table C.2: SA demand nodal mapping.

SA Grid Node	**Percentage of Total Demand Trace**
Adelaide Metro 275	50.67%
Eyre Peninsula 132	2.83%
Iron Triangle 275	8.65%
Upper North 275	0.10%
Mid North 330	23.38%
Riverland 132	5.68%
South East 275	8.68%

Appendix C.4. Transmission Line Parameters

Currently, there is no publicly available data for transmission line resistance and reactance parameters. Therefore, the parameters of the overhead transmission line are estimated in Table C.3. Per unit figures are given on $S_{base} = 10GVA$ and the respective voltage.

To determine the relevant properties of the system branches, ElectraNet's latest transmission line ratings, in conjunction with the Digital Atlas of Australia Electricity Transmission dataset and Table C.3 can be used. Tables C.4–C.16 below depict these parameters. Note that where publicly available information is available for future transmission projects, these have been included. Otherwise, a conservative 10% growth factor is applied to the sum of the highest voltage transmission lines in a given region.

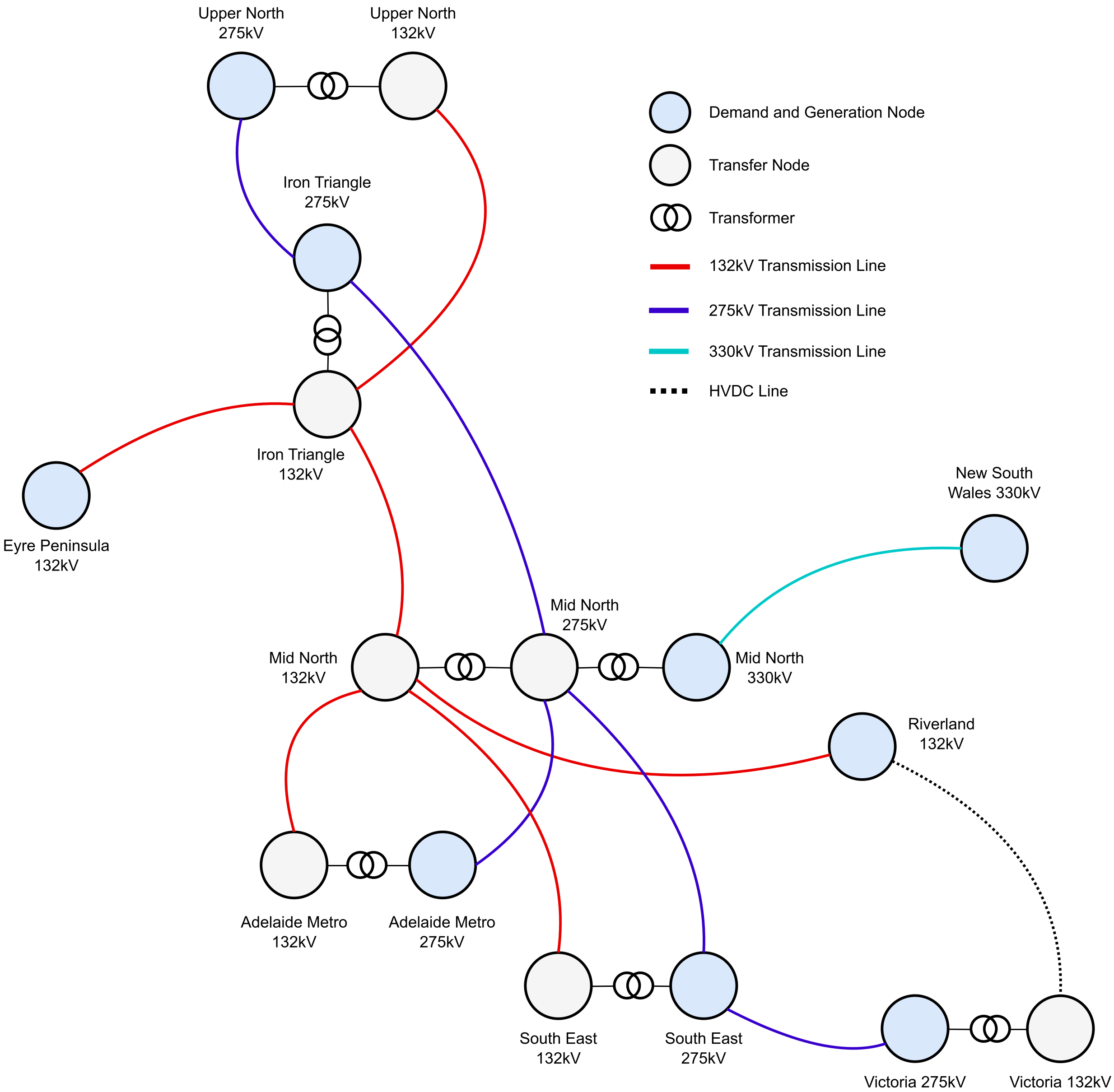

Figure C.1: Nodal representation of the South Australian electricity grid.

Table C.3: Overhead transmission line parameters (50Hz).

Voltage (kV)	**Resistance** (Ω/km)	**Reactance** (Ω/km)	**Resistance** (pu/km)	**Reactance** (pu/km)
132	0.112	0.39	0.0643	0.2238
275	0.04	0.319	0.0053	0.0422
330	0.03	0.3	0.0028	0.0275

Table C.4: Adelaide Metro to Mid North 132kV transmission lines.

Adelaide Metro Substation	**Mid North Substation**	**ElectraNet Feeder ID**	**Summer Rating (MW)**	**Winter Rating (MW)**	**Length (km)**	**Resistance (pu)**	**Reactance (pu)**
Cherry Gardens	Mount Barker	F1825	181	223	20.1	1.2909	4.4951
Total Equivalent			**181**	**223**		**1.2909**	**4.4951**

Table C.5: Adelaide Metro to Mid North 275kV transmission lines.

Adelaide Metro Substation	**Mid North Substation**	**ElectraNet Feeder ID**	**Summer Rating (MW)**	**Winter Rating (MW)**	**Length (km)**	**Resistance (pu)**	**Reactance (pu)**
Torrens Island Power Station	Para	F1902	857	857	21.0	0.1110	0.8850
Parafield Gardens West	Para	F1940	917	1139	12.84	0.0679	0.5416
Magill	Para	F1913	600	682	23.0	0.1217	0.9705
Cherry Gardens	Mount Barker South	F1955	909	1128	19.8	0.1049	0.8363
Cherry Gardens	Tungkillo	F1944	1188	1351	58.1	0.3071	2.4494
Adelaide Metro Substation	*Bundey*		*2,000* [85]	*2,000* [85]	*115* [61]	*0.6083*	*4.8509*
Total Equivalent			**6,471**	**7,157**		**0.0216**	**0.1719**

Table C.6: Eyre Peninsula to Iron Triangle 132kV transmission lines.

Eyre Peninsula Substation	**Iron Triangle Substation**	**ElectraNet Feeder ID**	**Summer Rating (MW)**	**Winter Rating (MW)**	**Length (km)**	**Resistance (pu)**	**Reactance (pu)**
Yadnarie #1	Cultana	F1882	241	273	136.4	8.7691	30.5354
Yadnarie #2	Cultana	F1883	291	331	143.6	9.2296	32.1389
Growth Factor			53	60			
Total Equivalent			**585**	**664**		**4.4967**	**15.6583**

Table C.7: Iron Triangle to Upper North 132kV transmission lines.

Iron Triangle Substation	**Upper North Substation**	**ElectraNet Feeder ID**	**Summer Rating (MW)**	**Winter Rating (MW)**	**Length (km)**	**Resistance (pu)**	**Reactance (pu)**
Davenport	Mount Gunson Tee	F1812	60	97	138.9	8.9288	31.0912
Davenport	Leigh Creek	F1813	17	61	86.2	5.5440	19.3050
Total Equivalent			**77**	**158**		**3.4203**	**11.9099**

Table C.8: Iron Triangle to Upper North 275kV transmission lines.

Iron Triangle Substation	**Upper North Substation**	**ElectraNet Feeder ID**	**Summer Rating (MW)**	**Winter Rating (MW)**	**Length (km)**	**Resistance (pu)**	**Reactance (pu)**
Davenport	Mount Gunson South	F1974	381	381	143.4	0.7583	6.0476
Growth Factor			38	38			
Total Equivalent			**419**	**419**		**0.7583**	**6.0476**

Table C.9: Iron Triangle to Mid North 132kV transmission lines.

Iron Triangle Substation	**Mid North Substation**	**ElectraNet Feeder ID**	**Summer Rating (MW)**	**Winter Rating (MW)**	**Length (km)**	**Resistance (pu)**	**Reactance (pu)**
Baroota	Bungama	F1856	11	76	26.3	1.6896	5.8835
Total Equivalent			**11**	**76**		**1.6896**	**5.8835**

Table C.10: Iron Triangle to Mid North 275kV transmission lines.

Iron Triangle Substation	**Mid North Substation**	**ElectraNet Feeder ID**	**Summer Rating (MW)**	**Winter Rating (MW)**	**Length (km)**	**Resistance (pu)**	**Reactance (pu)**
Davenport	Brinkworth	F1910	454	563	148.7	0.7865	6.2725
Davenport	Belalie	F1919, F1920	594	675	129.3	0.6839	5.4542
Davenport	Bungama	F1941	454	563	90.4	0.4780	3.8124
*Cultana**	*Bundey**	*1,100* [86]	*1,100* [86]	*250***	*1.3223*	*10.5455*	
Total Equivalent			**2,602**	**2,901**		**0.1792**	**1.4288**

*ElectraNet currently exploring 275kV, 330kV and 550kV options for this transmission project [86]. **Estimated distance.

Table C.11: Mid North to Riverland 132kV transmission lines.

Mid North Substation	**Riverland Substation**	**ElectraNet Feeder ID**	**Summer Rating (MW)**	**Winter Rating (MW)**	**Length (km)**	**Resistance (pu)**	**Reactance (pu)**
Robertstown	North West Bend	F1846	134	150	60.4	3.8805	13.5216
Morgan-Whyalla Pipeline 1	Morgan-Whyalla Pipeline 2	F1853	183	206	25.1	1.6131	5.6171
Growth Factor			*32*	*36*			
Total Equivalent			**349**	**392**		**1.1395**	**3.9677**

Table C.12: Mid North to South East 132kV transmission lines.

Mid North Substation	South East Substation	ElectraNet Feeder ID	Summer Rating (MW)	Winter Rating (MW)	Length (km)	Resistance (pu)	Reactance (pu)
Mobilong	Tailem Bend	F1835	182	182	32.4	2.0844	7.2581
Total Equivalent			**182**	**182**		**2.0844**	**7.2581**

Table C.13: Mid North to South East 275kV transmission lines.

Mid North Substation	South East Substation	ElectraNet Feeder ID	Summer Rating (MW)	Winter Rating (MW)	Length (km)	Resistance (pu)	Reactance (pu)
Tungkillo #1	Tailem Bend	F1835	458	569	123.9	0.6552	5.2250
Tungkillo #2	Tailem Bend	F1835	594	675	62.2	0.3292	2.6254
Growth Factor			*105*	*124*			
Total Equivalent			**1,157**	**1,368**		**0.2191**	**1.7474**

Table C.14: Riverland to Victoria 150kV* transmission lines.

Riverland Substation	Victorian Substation	ElectraNet Feeder ID	Summer Rating (MW)	Winter Rating (MW)	Length (km)	Resistance (pu)	Reactance (pu)
Monash	Red Cliffs		220	220	174.4	??	0
Total Equivalent			**220**	**220**		**???**	**0**

*Approximated for the model as 132kV. Modelled separately to the AC transmission lines as this line has 0 reactance and thus does not work with DC OPF. Resistance calculated assuming $1000m^2$ copper cable with resistance of $22m\Omega/km$ [87].

Table C.15: South East to Victoria 275kV transmission lines.

South East Substation	Victorian Substation	ElectraNet Feeder ID	Summer Rating (MW)	Winter Rating (MW)	Length (km)	Resistance (pu)	Reactance (pu)
South East	Heywood	F1930, F1931	594	675	91.0	2.0886	16.6563
Total Equivalent			**594**	**675**		**2.0886**	**16.6563**

Table C.16: Mid North to NSW 330kV transmission lines.

Mid North Substation	**NSW Substation**	**ElectraNet Feeder ID**	**Summer Rating (MW)**	**Winter Rating (MW)**	**Length (km)**	**Resistance (pu)**	**Reactance (pu)**
Bundey	Buronga	F1930, F1931	800	800	341.0	0.9394	9.3939
Total Equivalent			**800**	**800**		**0.9394**	**9.3939**

Nodes at different voltages in the same region, as defined in Figure C.1 are connected by transformers. A simplified model is presented such that the connected of these nodes are facilitated by a single transformer. The reactance the transformer connecting nodes i and j. The reactance is based on the summer rating as this is more conservative. However, the transformers are given both a winter and summer rating. Note that the resistance of these transformer is assumed to be 0.

$$X_{ij} = 0.12 \frac{S_{base}}{P^{\max}_{ij,summer}} \tag{C.1}$$

These transformers are rated the same as the high voltage line so as to not cause congestion in the model. The table below depicts the parameters of each transformer.

Table C.17: Transformer parameters.

Low Voltage Node	**High Voltage Node**	**Summer Rating (MW)**	**Winter Rating (MW)**	**Reactance (pu)**
Adelaide Metro 132kV	Adelaide Metro 275kV	6,471	7,157	0.1854
Iron Triangle 132kV	Iron Triangle 275kV	2,602	2,901	0.4612
Upper North 132kV	Upper North 275kV	419	419	2.8640
Mid North 132kV	Mid North 275kV	6,471	7,157	0.1854
Mid North 275kV	Mid North 330kV	6,471	7,157	0.1854
South East 132kV	South East 275kV	1,157	1,368	1.0372
Victoria 132kV	Victoria 275kV	594	675	5.4545

Combining the above tables together results in the following matrices.

Table C.18: Branch summer ratings (MW).

Node	AM 132	AM 275	EP 132	IT 132	IT 275	UN 132	UN 275	MN 132	MN 275	MN 330	R 132	SE 132	SE 275	VIC 132	VIC 275	NSW 330
Adelaide Metro 132	0	6,471	0	0	0	0	0	181	0	0	0	0	0	0	0	0
Adelaide Metro 275	6,471	0	0	0	0	0	0	0	6,471	0	0	0	0	0	0	0
Eyre Peninsula 132	0	0	0	585	0	0	0	0	0	0	0	0	0	0	0	0
Iron Triangle 132	0	0	585	0	2,602	77	0	11	0	0	0	0	0	0	0	0
Iron Triangle 275	0	0	0	2,602	0	0	419	0	2,602	0	0	0	0	0	0	0
Upper North 132	0	0	0	77	0	0	419	0	0	0	0	0	0	0	0	0
Upper North 275	0	0	0	0	419	419	0	0	0	0	0	0	0	0	0	0
Mid North 132	181	0	0	11	0	0	0	0	6,471	0	349	182	0	0	0	0
Mid North 275	0	6,471	0	0	2,602	0	0	6,471	0	6,471	0	0	1,157	0	0	0
Mid North 330	0	0	0	0	0	0	0	0	6,471	0	0	0	0	0	0	800
Riverland 132	0	0	0	0	0	0	0	349	0	0	0	0	0	220	0	0
South East 132	0	0	0	0	0	0	0	182	0	0	0	0	1,157	0	0	0
South East 275	0	0	0	0	0	0	0	0	1,157	0	0	1,157	0	0	594	0
VIC 132	0	0	0	0	0	0	0	0	0	0	220	0	0	0	220	0
VIC 275	0	0	0	0	0	0	0	0	0	0	0	0	594	220	0	0
NSW 330	0	0	0	0	0	0	0	0	0	800	0	0	0	0	0	0

Table C.19: Branch winter ratings (MW).

Node	AM 132	AM 275	EP 132	IT 132	IT 275	UN 132	UN 275	MN 132	MN 275	MN 330	R 132	SE 132	SE 275	VIC 132	VIC 275	NSW 330
Adelaide Metro 132	0	7,157	0	0	0	0	0	223	0	0	0	0	0	0	0	0
Adelaide Metro 275	7,157	0	0	0	0	0	0	0	7,157	0	0	0	0	0	0	0
Eyre Peninsula 132	0	0	0	664	0	0	0	0	0	0	0	0	0	0	0	0
Iron Triangle 132	0	0	664	0	2,901	158	0	76	0	0	0	0	0	0	0	0
Iron Triangle 275	0	0	0	2,901	0	0	419	0	2,901	0	0	0	0	0	0	0
Upper North 132	0	0	0	158	0	0	419	0	0	0	0	0	0	0	0	0
Upper North 275	0	0	0	0	419	419	0	0	0	0	0	0	0	0	0	0
Mid North 132	223	0	0	76	0	0	0	0	7,157	0	392	182	0	0	0	0
Mid North 275	0	7,157	0	0	2,901	0	0	7,157	0	7,157	0	0	1,368	0	0	0
Mid North 330	0	0	0	0	0	0	0	0	7,157	0	0	0	0	0	0	800
Riverland 132	0	0	0	0	0	0	0	392	0	0	0	0	0	220	0	0
South East 132	0	0	0	0	0	0	0	182	0	0	0	0	1,368	0	0	0
South East 275	0	0	0	0	0	0	0	0	1,368	0	0	1,368	0	0	675	0
VIC 132	0	0	0	0	0	0	0	0	0	0	220	0	0	0	675	0
VIC 275	0	0	0	0	0	0	0	0	0	0	0	0	675	675	0	0
NSW 330	0	0	0	0	0	0	0	0	0	800	0	0	0	0	0	0

Table C.20: Branch reactance matrix (pu).

Node	AM 132	AM 275	EP 132	IT 132	IT 275	UN 132	UN 275	MN 132	MN 275	MN 330	R 132	SE 132	SE 275	VIC 132	VIC 275	NSW 330
Adelaide Metro 132	0	0.1854	0	0	0	0	0	4.4951	0	0	0	0	0	0	0	0
Adelaide Metro 275	0.1854	0	0	0	0	0	0	0	0.1719	0	0	0	0	0	0	0
Eyre Peninsula 132	0	0	0	15.6583	0	0	0	0	0	0	0	0	0	0	0	0
Iron Triangle 132	0	0	15.6583	0	0.4612	11.9099	0	5.8835	0	0	0	0	0	0	0	0
Iron Triangle 275	0	0	0	0.4612	0	0	6.0476	0	1.4288	0	0	0	0	0	0	0
Upper North 132	0	0	0	11.9099	0	0	2.864	0	0	0	0	0	0	0	0	0
Upper North 275	0	0	0	0	6.0476	2.864	0	0	0	0	0	0	0	0	0	0
Mid North 132	4.4951	0	0	5.8835	0	0	0	0	0.1854	0	3.9677	7.2581	0	0	0	0
Mid North 275	0	0.1719	0	0	1.4288	0	0	0.1854	0	0.1854	0	0	1.7474	0	0	0
Mid North 330	0	0	0	0	0	0	0	0	0.1854	0	0	0	0	0	0	9.3939
Riverland 132	0	0	0	0	0	0	0	3.9677	0	0	0	0	0	0*	0	0
South East 132	0	0	0	0	0	0	0	7.2581	0	0	0	0	1.0372	0	0	0
South East 275	0	0	0	0	0	0	0	0	1.7474	0	0	1.0372	0	0	16.6563	0
VIC 132	0	0	0	0	0	0	0	0	0	0	0*	0	0	0	5.4545	0
VIC 275	0	0	0	0	0	0	0	0	0	0	0	0	16.6563	5.4545	0	0
NSW 330	0	0	0	0	0	0	0	0	0	9.3939	0	0	0	0	0	0

*Denotes a high voltage DC line.

Table C.21: Branch resistance matrix (pu).

Node	AM 132	AM 275	EP 132	IT 132	IT 275	UN 132	UN 275	MN 132	MN 275	MN 330	R 132	SE 132	SE 275	VIC 132	VIC 275	NSW 330
Adelaide Metro 132	0	0*	0	0	0	0	0	1.2909	0	0	0	0	0	0	0	0
Adelaide Metro 275	0*	0	0	0	0	0	0	0	0.0223	0	0	0	0	0	0	0
Eyre Peninsula 132	0	0	0	4.4967	0	0	0	0	0	0	0	0	0	0	0	0
Iron Triangle 132	0	0	4.4967	0	0*	3.4203	0	1.6896	0	0	0	0	0	0	0	0
Iron Triangle 275	0	0	0	0*	0	0	0.7583	0	0.2072	0	0	0	0	0	0	0
Upper North 132	0	0	0	3.4203	0	0	0*	0	0	0	0	0	0	0	0	0
Upper North 275	0	0	0	0	0.7583	0*	0	0	0	0	0	0	0	0	0	0
Mid North 132	1.2909	0	0	1.6896	0	0	0	0	0*	0	1.1395	2.0844	0	0	0	0
Mid North 275	0	0.0223	0	0	0.2072	0	0	0*	0	0*	0	0	0.2191	0	0	0
Mid North 330	0	0	0	0	0	0	0	0	0*	0	0	0	0	0	0	0.9394
Riverland 132	0	0	0	0	0	0	0	1.1395	0	0	0	0	0	11.2099	0	0
South East 132	0	0	0	0	0	0	0	2.0844	0	0	0	0	0*	0	0	0
South East 275	0	0	0	0	0	0	0	0	0.2191	0	0	0*	0	0	2.0886	0
VIC 132	0	0	0	0	0	0	0	0	0	0	11.2099	0	0	0	0*	0
VIC 275	0	0	0	0	0	0	0	0	0	0	0	0	2.0886	0*	0	0
NSW 330	0	0	0	0	0	0	0	0	0	0.9394	0	0	0	0	0	0

*Denotes a branch that exists but with 0 resistance (transformer branches).

Appendix C.5. Demand Side Participation

The power and energy limits for DSP in the investigation period are highlighted in Table C.22.

Table C.22: DSP parameters.

DSP Component	**2045**	**2046**	**2047**	**2048**	**2049**	**2050)**	**2051**
DSP Power (MW)	182.0	187.0	192.2	197.3	202.6	206.1	207.7
DSP Energy (MWh)	69,091	89,382	117,820	116,054	123,772	192,871	224,422

Appendix D. Base Case Detailed Results

The detailed results of the base case multi-nodal grid model are given in Table D.1.

Table D.1: Summary of the multi-nodal grid base case results.

Node	**Technology**	**Component**	**Class**	**2045**	**2046**	**2047**	**2048**	**2049**	**2050**	**2051**
Adelaide Metro 275	Lithium	Power (GW)	Shallow	0.2798	0	0	0	0	0	0
			Medium	0.4909	0.5965	1.110	0.5340	1.241	1.215	1.802
			Deep	1.240	1.134	0.8468	1.323	1.257	1.040	1.486
		Energy (GWh)	Shallow	1.119	0	0	0	0	0	0
			Medium	5.891	7.157	13.32	6.408	14.89	14.58	21.62
			Deep	264.8	123.3	134.4	170.1	189.6	88.68	469.5
Eyre Peninsula 132	Lithium	Power (GW)	Shallow	0.0216	0.0990	0	0	0.1315	0.1413	0.0512
			Medium	0.0835	0.1841	0.1232	0.1586	0.0907	0.0244	0.0877
			Deep	0	0	0.0311	0.0091	0.0091	0.0510	0.0391
		Energy (GWh)	Shallow	0.0863	0.3961	0	0	0.5259	0.5654	0.2048
			Medium	1.002	2.210	1.478	1.903	1.088	0.2932	1.052
			Deep	0	0	9.088	0.7971	1.495	5.980	15.05
Mid North 330	Lithium	Power (GW)	Shallow	0	0	0	0	0	0	0
			Medium	0.1103	0	0	0.1505	0.3411	0.0157	0
			Deep	0	0	0	0	0	0	0.3017
		Energy (GWh)	Shallow	0	0	0	0	0	0	0
			Medium	1.323	0	0	1.806	4.093	0.1881	0
			Deep	0	0	0	0	0	0	13.83
Riverland 132	Lithium	Power (GW)	Shallow	0.1002	0	0.2110	0.0540	0.1292	0.0314	0
			Medium	0.5257	0.7039	0.5986	0.5982	0.6566	0.7378	0.5212
			Deep	0.3274	0	0.0979	0.1158	0.0124	0	0.4885
		Energy (GWh)	Shallow	0.4006	0	0.8442	0.2159	0.5167	0.1255	0
			Medium	6.308	8.447	7.183	7.178	7.879	8.853	6.255
			Deep	99.80	0	23.38	15.86	1.194	0	15.22
South East 275	Lithium	Power (GW)	Shallow	0	0	0	0	0	0	0
			Medium	0.2545	0.0740	0.4001	0.1400	0.1792	0.1377	0.7314
			Deep	0.1834	0.1059	0.0574	0.1807	0.2109	0.1110	0.1258
		Energy (GWh)	Shallow	0	0	0	0	0	0	0
			Medium	3.054	0.8875	4.801	1.680	2.151	1.653	8.777
			Deep	24.75	10.54	3.906	23.72	13.68	7.373	17.36
Iron Triangle 275	Lithium	Power (GW)	Shallow	0.2724	0.3201	0.3461	0.2139	0.1912	0.1036	0.2224

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Node	**Technology**	**Component**	**Class**	**2045**	**2046**	**2047**	**2048**	**2049**	**2050**	**2051**
			Medium	0.8763	0.8475	0.8599	0.9506	0.8449	0.9994	0.8954
			Deep	0	0	0	0	0	0	0
		Energy (GWh)	Shallow	1.090	1.280	1.384	0.8556	0.7648	0.4145	0.8895
			Medium	7.090	10.06	10.32	10.19	10.14	11.99	9.626
			Deep	0	0	0	0	0	0	0
	PHES	Power (GW)	Medium	0	0	0	0	0	0	0
			Deep	0	0.1712	0.1393	0	0.1787	0.2524	0.1453
		Energy (GWh)	Medium	0	0	0	0	0	0	0
			Deep	0	34.01	62.4	0	56.69	62.4	62.4
Upper North 275	Lithium	Power (GW)	Shallow	0.0373	0	0.0757	0	0.1211	0.0342	0.0586
			Medium	0.2810	0.0120	0.2013	0.2705	0.1795	0	0.3222
			Deep	0	0	0	0	0	0	0
		Energy (GWh)	Shallow	0.1491	0	0.3027	0	0.4845	0.1368	0.2344
			Medium	1.998	0.0834	2.415	1.860	2.154	0	2.289
			Deep	0	0	0	0	0	0	0
Total		**Power (GW)**		**5.084**	**4.248**	**5.099**	**4.699**	**5.774**	**4.895**	**7.278**
		Energy (GWh)		**418.8**	**198.4**	**275.3**	**242.6**	**307.4**	**203.2**	**644.2**

The annual renewable energy generation and demand for each year is shown in Table D.2.

Table D.2: Annual generation, demand, and storage requirements for the base case scenario.

Year	**Solar (TWh)**	**Wind (TWh)**	**Variable Renewable (TWh)**	**Demand (TWh)**	**Generation Exceedance (%)**	**Storage Requirement (GWh)**
2045	12.29	14.05	26.35	18.57	42%	418.8
2046	13.14	16.66	29.81	18.97	57%	198.4
2047	12.93	17.35	30.28	19.07	59%	275.3
2048	13.09	16.74	29.83	19.00	57%	242.6
2049	12.81	16.44	29.24	19.44	50%	307.4
2050	12.57	18.18	30.76	19.37	59%	203.2
2051	12.80	16.77	29.57	19.05	55%	644.2

Appendix E. Dispatchable Generator Detailed Results

Detailed dispatchable generator results across the investigation period are outlined in Tables E.1–E.7 below.

Table E.1: Summary of the 2045 storage requirements for different generator dispatch sizes and percentages of demand met.

Dispatch Generator Size (GW)	Max 1% Annual Demand			Max 2% Annual Demand			Max 3% Annual Demand		
	Power (GW)	Storage (GWh)	Cost ($B)	Power (GW)	Storage (GWh)	Cost ($B)	Power (GW)	Storage (GWh)	Cost ($B)
0.2	4.842	256.1	61.55	4.842	256.1	61.55	4.842	256.1	61.55
0.4	5.039	220.3	56.25	3.810	215.4	51.40	3.810	215.4	51.40
0.6	5.073	220.8	56.03	3.750	173.9	44.00	3.612	176.8	43.28
0.8	5.075	221.1	56.02	4.349	152.3	42.08	3.471	136.3	35.89
1.0	5.075	221.1	56.02	4.399	151.3	42.02	4.088	106.3	32.41

Table E.2: Summary of the 2046 storage requirements for different generator dispatch sizes and percentages of demand met.

Dispatch Generator Size (GW)	Max 1% Annual Demand			Max 2% Annual Demand			Max 3% Annual Demand		
	Power (GW)	Storage (GWh)	Cost ($B)	Power (GW)	Storage (GWh)	Cost ($B)	Power (GW)	Storage (GWh)	Cost ($B)
0.2	4.271	149.1	39.62	4.271	149.1	39.62	4.271	149.1	39.62
0.4	3.365	115.4	31.81	3.365	115.4	31.81	3.365	115.4	31.81
0.6	3.455	92.10	27.84	3.054	94.62	26.94	3.054	94.62	26.94
0.8	3.712	87.69	27.60	3.169	73.94	23.15	2.873	76.66	22.67
1.0	3.717	87.62	27.59	3.599	66.67	22.47	3.075	56.43	18.97

Table E.3: Summary of the 2047 storage requirements for different generator dispatch sizes and percentages of demand met.

Dispatch Generator Size (GW)	Max 1% Annual Demand			Max 2% Annual Demand			Max 3% Annual Demand		
	Power (GW)	Storage (GWh)	Cost ($B)	Power (GW)	Storage (GWh)	Cost ($B)	Power (GW)	Storage (GWh)	Cost ($B)
0.2	4.257	117.8	34.22	4.258	117.9	34.22	4.258	117.9	34.22
0.4	4.190	101.4	31.54	3.641	85.37	26.80	3.641	85.37	26.80
0.6	4.188	101.4	31.52	3.710	74.20	24.34	3.257	71.18	22.84
0.8	4.188	101.4	31.52	3.714	73.99	24.27	3.119	59.29	19.81
1.0	4.188	101.4	31.52	3.713	73.95	24.27	3.391	55.37	19.48

Table E.4: Summary of the 2048 storage requirements for different generator dispatch sizes and percentages of demand met.

Dispatch Generator Size (GW)	Max 1% Annual Demand			Max 2% Annual Demand			Max 3% Annual Demand		
	Power (GW)	Storage (GWh)	Cost ($B)	Power (GW)	Storage (GWh)	Cost ($B)	Power (GW)	Storage (GWh)	Cost ($B)
0.2	4.167	204.2	51.46	4.167	204.2	51.46	4.167	204.2	51.46
0.4	3.757	172.7	44.34	3.757	172.7	44.34	3.757	172.7	44.34
0.6	3.546	140.7	37.47	3.546	140.7	37.47	3.546	140.7	37.47
0.8	4.102	106.2	31.98	3.344	109.0	30.65	3.344	109.0	30.65
1.0	4.706	92.76	30.87	3.343	77.53	24.39	3.125	78.28	23.94

Table E.5: Summary of the 2049 storage requirements for different generator dispatch sizes and percentages of demand met.

Dispatch Generator Size (GW)	Max 1% Annual Demand			Max 2% Annual Demand			Max 3% Annual Demand		
	Power (GW)	Storage (GWh)	Cost ($B)	Power (GW)	Storage (GWh)	Cost ($B)	Power (GW)	Storage (GWh)	Cost ($B)
0.2	5.650	192.3	53.17	5.650	192.3	53.17	5.650	192.3	53.17
0.4	4.841	131.9	41.56	4.465	134.4	41.08	4.465	134.4	41.08
0.6	4.955	132.3	40.60	4.192	106.5	32.97	4.192	106.5	32.97
0.8	4.955	132.3	40.60	4.236	93.71	30.29	3.796	81.01	25.69
1.0	4.955	132.3	40.60	4.191	93.79	30.23	3.977	71.30	24.47

Table E.6: Summary of the 2050 storage requirements for different generator dispatch sizes and percentages of demand met.

Dispatch Generator Size (GW)	Max 1% Annual Demand			Max 2% Annual Demand			Max 3% Annual Demand		
	Power (GW)	Storage (GWh)	Cost ($B)	Power (GW)	Storage (GWh)	Cost ($B)	Power (GW)	Storage (GWh)	Cost ($B)
0.2	4.903	127.8	37.85	4.903	127.8	37.85	4.903	127.8	37.85
0.4	3.645	96.96	30.41	3.445	98.68	30.13	3.445	98.68	30.13
0.6	3.938	90.34	29.08	3.121	86.49	26.67	3.121	86.49	26.67
0.8	4.142	88.49	28.81	3.268	71.74	23.81	2.939	74.04	23.31
1.0	4.152	88.26	28.78	3.804	65.65	22.78	2.935	60.57	20.24

Table E.7: Summary of the 2051 storage requirements for different generator dispatch sizes and percentages of demand met.

Dispatch Generator Size (GW)	Max 1% Annual Demand			Max 2% Annual Demand			Max 3% Annual Demand		
	Power (GW)	Storage (GWh)	Cost ($B)	Power (GW)	Storage (GWh)	Cost ($B)	Power (GW)	Storage (GWh)	Cost ($B)
0.2	5.543	502.7	107.8	5.051	504.3	105.9	5.051	504.3	105.9
0.4	7.276	439.4	102.5	5.242	363.7	80.31	5.242	363.7	80.31
0.6	7.276	439.4	102.5	7.052	248.1	68.44	5.295	238.4	57.28
0.8	7.276	439.4	102.5	7.222	231.2	66.40	5.304	147.8	43.04
1.0	7.276	439.4	102.5	7.235	230.8	66.24	5.241	142.4	41.77

Appendix F. Weather Years Detailed Results

In Table F.1, the detailed results obtained from the 13 reference weather year analysis are provided, whereas Table F.2 provides the detailed results for the 30 block bootstrap years.

Table F.1: Problem results using AEMO's 13 years of weather data.

Weather Year	**Total Power (GW)**	**Total Storage (GWh)**	**BESS Power (GW)**	**BESS Storage (GWh)**	**PHES Power (GW)**	**PHES Storage (GWh)**	**Renewable Generation (TWh)**	**CAPEX (\$B)**	**OPEX (\$B)**	**System Cost (\$B)**
Year 1	4.756	343.8	4.465	281.4	0.291	62.4	31.44	72.11	3.734	75.84
Year 2	6.030	497.3	5.897	434.9	0.133	62.4	30.61	102.0	4.406	106.4
Year 3	4.724	177.7	4.689	173.7	0.035	4.06	30.36	44.05	3.552	47.60
Year 4	5.510	282.4	5.309	220.0	0.202	62.4	30.17	62.54	4.032	66.57
Year 5	5.763	677.9	5.633	615.5	0.130	62.4	28.03	136.3	5.244	141.6
Year 6	4.782	279.3	4.782	279.3	0	0	28.09	64.15	4.233	68.38
Year 7	5.845	333.8	5.639	271.4	0.206	62.4	30.00	73.23	4.564	77.79
Year 8	4.238	192.2	4.056	166.4	0.182	25.8	30.91	46.85	4.056	50.91
Year 9	4.365	213.6	4.125	186.7	0.239	26.9	29.73	49.52	3.543	53.07
Year 10	5.224	365.5	5.030	303.1	0.194	62.4	28.98	78.05	4.420	82.47
Year 11	5.094	159.3	4.897	125.4	0.198	34.0	30.86	39.93	3.519	43.45
Year 12	5.495	617.4	5.319	555.0	0.175	62.4	29.47	122.8	4.178	127.0
Year 13	4.274	288.1	4.001	245.0	0.272	43.1	31.56	61.80	3.385	65.18

Table F.2: Problem results using 30 years of synthetic trace data.

Bootstrap Year	**Total Power (GW)**	**Total Storage (GWh)**	**BESS Power (GW)**	**BESS Storage (GWh)**	**PHES Power (GW)**	**PHES Storage (GWh)**	**CAPEX ($B)**	**OPEX ($B)**	**System Cost ($B)**
Year 1	4.182	215.6	4.040	205.6	0.1424	10.08	50.83	3.745	54.58
Year 2	4.499	334.2	4.245	304.4	0.2533	29.84	72.24	3.980	76.22
Year 3	4.702	320.5	4.481	258.9	0.2211	61.62	67.06	3.320	70.38
Year 4	4.236	182.1	4.105	169.2	0.1307	12.86	44.52	3.704	48.22
Year 5	6.189	262.1	6.095	231.2	0.0938	30.90	61.15	4.262	65.41
Year 6	4.284	117.9	4.060	99.8	0.2241	18.04	32.43	3.467	35.90
Year 7	4.277	214.1	4.277	214.1	0	0	50.63	3.667	54.29
Year 8	7.439	425.0	7.247	362.6	0.1920	62.40	95.40	6.422	101.8
Year 9	4.137	192.6	4.043	182.2	0.0937	10.44	46.32	3.627	49.95
Year 10	4.889	238.6	4.719	200.3	0.1702	38.29	55.16	4.098	59.26
Year 11	4.356	310.7	3.996	252.2	0.3600	58.48	66.27	3.729	70.00
Year 12	5.680	366.8	5.563	340.2	0.1176	26.58	82.87	5.551	88.42
Year 13	4.175	197.1	3.802	172.7	0.3735	24.35	46.83	3.622	50.45
Year 14	4.230	221.3	4.056	197.8	0.1738	23.49	52.03	4.022	56.05
Year 15	5.189	377.4	5.134	369.7	0.0548	7.66	80.63	3.805	84.44
Year 16	4.991	215.7	4.847	189.1	0.1434	26.66	50.76	3.758	54.52
Year 17	4.252	336.7	4.037	274.3	0.2154	62.40	70.78	3.828	74.61
Year 18	4.030	243.8	3.751	194.0	0.2792	49.83	54.74	3.926	58.66
Year 19	5.142	273.6	5.037	250.5	0.1050	23.03	61.67	3.891	65.56
Year 20	4.550	627.9	4.373	565.5	0.1773	62.40	124.0	4.197	128.1
Year 21	5.013	325.8	4.827	263.4	0.1865	62.40	69.63	3.938	73.57
Year 22	4.283	348.3	3.896	294.7	0.3867	53.57	74.99	4.480	79.47
Year 23	4.202	148.9	4.126	138.4	0.0767	10.52	37.98	3.400	41.38
Year 24	4.531	180.5	4.194	143.7	0.3367	36.83	43.78	3.698	47.48
Year 25	4.054	231.2	3.620	194.8	0.4343	36.41	53.24	3.913	57.15
Year 26	4.585	242.3	4.455	220.7	0.1304	21.59	54.00	3.140	57.14
Year 27	4.268	208.4	3.995	147.5	0.2728	60.91	46.72	3.367	50.09
Year 28	4.732	164.1	4.658	154.6	0.0741	9.489	41.07	3.400	44.46
Year 29	4.037	428.5	3.802	366.1	0.2354	62.40	87.04	3.784	90.82
Year 30	4.271	247.9	3.933	185.9	0.3381	62.02	53.97	3.408	57.38

Appendix G. Co-Optimisation Model

This appendix outlines the co-optimisation model formulation and the detailed results, respectively.

Appendix G.1. Formulation

To co-optimise both generation and storage, the optimisation problem needs to be extended. The following equations depicts the constraints that have been added or modified to incorporate this.

New Decision Variables

$P^{\max}_{VRE,r,g}$	Installed capacity of generation technology g at local node r (pu)
$P_{VRE,r,g,t}$	Power generation at local node r for generation technology g at time t (pu)

New Parameters

$C_{P,g}$	Power capacity cost for generation technology g (pu adjusted)
$C_{O,g}$	Present value of operating expenditure over the analysis period for generation technology g (pu adjusted)
$C_{A,O,g}$	Annual operating expenditure of generation technology g (pu adjusted)
$P^{lim}_{VRE,r,g}$	Build limit at local node r for generation technology g (pu)
$g_{r,g,t}$	Generation trace at local node r for generation technology g at time t, where $0 \leq g_{r,g,t} \leq 1$

The optimisation objective is to minimise the present value of the operating and capital expenditure of both storage and generation as shown below.

$$\min \left(\sum_{r\in\mathcal{R}} \left(\sum_{b\in\mathcal{B}} \sum_{h\in\mathcal{H}} \left(C_{E,b,h} E^{\max}_{r,b,h} + (C_{P,b,h} + C_{O,b,h}) P^{\max}_{r,b,h} \right) \right) + \sum_{g\in\mathcal{G}} (C_{P,g} + C_{O,g}) P^{\max}_{VRE,r,g} \right) \tag{G.1}$$

Additional generation production is incorporated into the nodal energy balance equation as depicted below.

$$G_{r,t} + P_{\text{disc},r,t} + P_{I,r,t} - P_{\text{chrg},r,t} - P_{\text{curt},r,t} + P_{\text{DSP},r,t} - D_{r,t} + \text{USE}_{r,t} - P_{loss,r,t} + P_{VRE,r,t} = 0 \quad \forall r \in \mathcal{R}, \forall t \in \mathcal{T}, \tag{G.2}$$

where $G_{\text{r,t}}$ is the sum of all currently developed utility-scale generation at node r,

$$G_{r,t} = \sum_{g\in\mathcal{G}(\text{r})} G_{r,g,t}, \tag{G.3}$$

$$G_{r,g,t} \geq 0, \quad \forall r \in \mathcal{R}, \forall t \in \mathcal{T}. \tag{G.4}$$

where $P_{VRE,r,t}$ is the sum of all potential utility scale generation at node r.

$$P_{VRE,r,t} = \sum_{g\in\mathcal{G}(\text{r})} P^{\max}_{VRE,r,g} g_{r,t} \tag{G.5}$$

$$g_{r,t} \in [0, 1], \quad \forall r \in \mathcal{R}, \forall t \in \mathcal{T} \tag{G.6}$$

The amount of renewable energy that can be developed within each node is subject to build limits and is captured in (G.7).

$$P_{VRE,r,g}^{\max} \leq P_{VRE,r,g}^{lim}, \quad \forall g \in \mathcal{G}, \forall r \in \mathcal{R} \tag{G.7}$$

The build limits for VRE development, as defined in [84], for each REZs is defined in Table G.1. These limits are then assigned to the defined SA nodes shown in G.2.

Table G.1: REZ build limits.

REZ	Mapped SA Node	Solar Build Limit (MW)	Wind Build Limit (MW)
S1 South East	South East 275	100	3,200
S2 Riverland	Riverland 132	4,000	1,400
S3 Mid North	Mid North 330	1,300	4,600
S4 Yorke Peninsula	Mid North 330	0	1,400
S5 Northern South Australia	Iron Triangle 275	5,000	2,360
S6 Roxby Downs	Upper North 275	3,400	0
S7 Eastern Eyre Peninsula	Eyre Peninsula 132	5,000	2,300
S8 Western Eyre Peninsula	Eyre Peninsula 132	4,000	1,500

Table G.2: SA VRE build limits.

SA Node	Solar Build Limit (MW)	Wind Build Limit (MW)
South East 275	100	3,200
Riverland 132	4,000	1,400
Mid North 330	1,300	6,000
Iron Triangle 275	5,000	2,360
Upper North 275	3,400	0
Eyre Peninsula 132	9,000	3,800

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Table G.3 shows the installed VRE capacities and Table G.4 presents the storage breakdown.

Table G.3: Installed renewable generation capacity of the co-optimisation problem.

Node	Technology	2045	2046	2047	2048	2049	2050	2051
Adelaide Metro 275	Solar (GW)	0	0	0	0	0	0	0
	Wind (GW)	0	0	0	0	0	0	0
Eyre Peninsula 132	Solar (GW)	0.7340	0.946	0.374	0.707	1.190	0.5428	1.902
	Wind (GW)	0	0	0.3506	0	0	0.3218	0
Iron Triangle 275	Solar (GW)	0	0	0	0	0	0	0
	Wind (GW)	0	0	0	0	0	0	0
Upper North 275	Solar (GW)	1.841	3.271	1.460	2.627	3.4	1.718	3.4
	Wind (GW)	0	0	0	0	0	0	0
Mid North 330	Solar (GW)	1.3	1.3	1.3	1.3	1.3	1.3	1.3
	Wind (GW)	2.568	1.425	1.979	4.516	4.076	1.912	3.396
Riverland 132	Solar (GW)	3.963	3.970	3.453	3.970	3.970	3.970	3.970
	Wind (GW)	0	0	0	0	0	0	0
South East 275	Solar (GW)	0	0	0	0	0	0.013	0.013
	Wind (GW)	3.2	2.553	2.341	3.2	3.2	2.381	3.2

Table G.4: Summary of the co-optimisation problem storage results.

Node	**Technology**	**Component**	**Class**	**2045**	**2046**	**2047**	**2048**	**2049**	**2050**	**2051**
Adelaide Metro 275	Lithium	Power (GW)	Shallow	0.5403	0.0678	0	0	0	0	0
			Medium	0.2601	0.8896	0.9338	0.2523	1.655	0.9598	0.6098
			Deep	1.157	0.7722	0.9213	1.114	0.3714	0.5979	1.159
		Energy (GWh)	Shallow	1.739	0.2571	0	0	0	0	0
			Medium	3.121	10.68	11.21	3.027	19.86	11.52	7.32
			Deep	143.9	57.16	61.78	97.22	22.80	39.34	93.40
Eyre Peninsula 132	Lithium	Power (GW)	Shallow	0.0166	0.0886	0.0062	0	0.0569	0	0.0046
			Medium	0.1424	0.1239	0.1176	0.1711	0.1554	0.1138	0.2000
			Deep	0	0	0	0	0	0	0
		Energy (GWh)	Shallow	0.0663	0.3545	0.0250	0	0.2276	0	0.0186
			Medium	1.709	1.195	1.344	1.317	1.864	1.365	1.834
			Deep	0	0	0	0	0	0	0
Mid North 330	Lithium	Power (GW)	Shallow	0.0698	0	0	0	0	0	0
			Medium	0.2246	0	0.0690	0	0	0	0
			Deep	0	0	0	0	0	0	0
		Energy (GWh)	Shallow	0.2791	0	0	0	0	0	0
			Medium	2.695	0	0.8279	0	0	0	0
			Deep	0	0	0	0	0	0	0
Riverland 132	Lithium	Power (GW)	Shallow	0.3193	0.0676	0	0	0	0.6009	0.2832
			Medium	0.9661	1.264	1.288	1.209	1.439	0.7765	1.488
			Deep	0	0	0	0	0	0	0
		Energy (GWh)	Shallow	1.277	0.2705	0	0	0	2.349	1.133
			Medium	11.59	15.16	15.45	14.51	17.26	9.318	17.86
			Deep	0	0	0	0	0	0	0
South East 275	Lithium	Power (GW)	Shallow	0	0.1101	0	0.0292	0	0	0
			Medium	0.1537	0.0772	0.2072	0.3829	0.6981	0.2756	0.7481
			Deep	0.0567	0	0	0.0149	0	0.3127	0
		Energy (GWh)	Shallow	0	0.3277	0	0.1014	0	0	0
			Medium	1.845	0.9264	2.486	4.595	8.378	3.307	8.977
			Deep	5.641	0	0	1.238	0	17.40	0
The North 275	Lithium	Power (GW)	Shallow	0.0152	0	0	0	0	0	0

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Node	Technology	Component	Class	2045	2046	2047	2048	2049	2050	2051
			Medium	0.0234	0.1989	0.0131	0.1987	0.3897	0	0.2574
			Deep	0	0	0	0	0	0	0
		Energy (GWh)	Shallow	0.0610	0	0	0	0	0	0
			Medium	0.2808	2.3868	0.1568	2.385	4.676	0	3.089
			Deep	0	0	0	0	0	0	0
	PHES	Power (GW)	Medium	0	0	0	0	0	0	0
			Deep	0.0116	0.0359	0	0.0151	0.0003	0.3965	0.0714
		Energy (GWh)	Medium	0	0	0	0	0	0	0
			Deep	1.070	2.7847	0	1.084	0.0161	21.62	5.203
Upper North 275	Lithium	Power (GW)	Shallow	0.1493	0.0848	0.0633	0.0772	0.1705	0.0135	0.0014
			Medium	0.4801	1.189	0.4365	0.9189	1.106	0.5244	1.467
			Deep	0	0	0	0	0	0	0
		Energy (GWh)	Shallow	0.5973	0.3392	0.2531	0.3087	0.6819	0.0541	0.0057
			Medium	4.858	11.022	3.491	8.166	8.126	6.293	17.60
			Deep	0	0	0	0	0	0	0
Total		**Power (GW)**		**4.587**	**4.969**	**4.056**	**4.384**	**6.042**	**4.572**	**6.290**
		Energy (GWh)		**180.7**	**102.9**	**97.02**	**134.0**	**83.89**	**112.6**	**156.4**